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Integrative taxonomic revision of *Uropsilus* (Uropsilinae, Talpidae) combining genomic, morphological and geographic evidence

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ABSTRACT

Morphological conservatism and niche stasis frequently obscure species boundaries, especially in lineages that have experienced prolonged geographic isolation and restricted gene flow. The shrew moles of subfamily Uropsilinae, represented solely by the genus *Uropsilus*, exemplify this problem. Distributed mainly in mid- to high-elevation habitats of the Mountains of Southwestern China, *Uropsilus* currently includes 11 described species, and at least 18 evolutionarily independent lineages. The taxonomic status of multiple lineages remains uncertain, and new taxa continue to be proposed on an unstable systematic foundation. Here, we integrate genomic data with comprehensive morphological evidence to reassess species boundaries across the genus. We combined species delimitation incorporating speciation process, coalescent-based species trees with divergence time estimates, genome-wide patterns of absolute genetic divergence, and geographic distributions with both classical craniodental measurements and geometric morphometrics. We synthesized these lines of evidence within the context of a reticulate evolutionary. Our results indicate that, despite a complex history of introgression, long-term isolation has led to pronounced genomic divergence and stable morphological differentiation. Within an integrative taxonomic framework, we find consistent support for recognizing six lineages as distinct species, including five previously proposed putative species and one newly detected lineage, while one putative species is confirmed as conspecific with *U. gracilis*. We formally diagnose and describe these six species and provide an updated dichotomous key to all recognized members of the genus. Our study delivers a robust integrative framework for the taxonomy of this ancient montane clade and offers a methodological template for delimiting cryptic species in other rapidly diverged taxa.

1. Introduction

Species delimitation forms the cornerstone of biodiversity documentation and conservation, providing the essential framework for understanding speciation mechanisms and evolutionary processes (De Queiroz 2007; Wiens 2007). Despite its fundamental importance, delineating species boundaries remains one of the most challenging tasks in systematic biology, particularly when confronted with complex evolutionary phenomena such as adaptive radiation, ongoing gene flow, and clinal variation across geographic gradients (Irwin et al., 2001; Nosil 2008; Jackson et al., 2017).

The advent of genomic technologies has revolutionized the detection

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of fine-scale genetic differentiation and reconstruction of phylogenetic relationships with unprecedented resolution (Leache et al., 2014; Kapli et al., 2020), but it has also introduced new challenges for taxonomic practice. While genomic data can reveal previously undetected cryptic diversity (Herrera et al., 2022; Gu et al., 2023), the application of molecular delimitation methods without careful consideration of biological context (e.g., morphology, ecology, geographic distribution, and reproductive compatibility) may lead to taxonomic inflation through over-splitting of populations (Sukumaran and Knowles 2017). This problem is particularly acute in cases of recent vicariance, where interrupted gene flow may exaggerate divergence between geographically isolated populations (Chambers and Hillis 2019). On the other hand, reliance on traditional morphological approaches alone risks underestimating true species diversity, especially within cryptic species complexes, when pronounced genetic divergence is not accompanied by apparent morphological or ecological differentiation (Bickford et al., 2007; Pfenninger and Schwenk 2007; Struck et al., 2018). Despite the increase in genomic-based species delimitation studies, only a few have led to formal taxonomic revisions (Singhal et al., 2025). This gap underscores the importance of integrative taxonomy, which combines genomic data with morphological, ecological, and biogeographic evidence to enable robust species delimitation in taxonomically challenging groups (Dayrat 2005; Schlick-Steiner et al., 2009; Padial et al., 2010; Karbstein et al., 2024).

The genus *Uropsilus* Milne-Edwards, 1871 (Talpidae: Uropsilinae) exemplifies these taxonomic challenges. Mainly distributed in the Mountains of Southwest China (MSC) biodiversity hotspot and inhabiting moist forests in mid- to high-elevation habits (800–4000 m above sea level), this morphologically conserved group has been taxonomically contentious for over a century (see Supplementary Table S1). Early molecular phylogenetic studies revealed substantial cryptic diversity within *Uropsilus* (Wan et al., 2013; Wan et al., 2018), and subsequent work has shown that rapid radiation combined with ancient gene flow can generate conflicting phylogenetic signals across the genome, rendering species boundaries unstable and challenging to define (Li et al., 2026). Moreover, many molecularly divergent lineages identified to date lack clear, taxonomically stable morphological diagnostic features, preventing their formal recognition. Consequently, even with several new species described in recent years (Hu et al., 2021; Bui et al., 2023; Ren et al., 2023; Jiang et al., 2025), *Uropsilus* remains without a comprehensive and stable taxonomic framework. Its species diversity is likely still underestimated, posing challenges for subsequent evolutionary, biogeographic, and conservation studies.

Currently, *Uropsilus* comprises 11 formally described species: *U. soricipes* Milne-Edwards, 1871; *U. andersoni* Thomas, 1911; *U. gracilis* Thomas, 1911; *U. investigator* Thomas, 1922; *U. atronates* Allen, 1923; *U. nivatus* Allen, 1923; *U. aequodonenia* Liu et al., 2013; *U. dabieshanensis* (Hu et al., 2021); *U. fansipanensis* (Bui et al., 2023); *U. huanggangensis* Chen, Jiang & Ren, 2023; and *U. funiushanensis* Jiang & Lu, 2023. At least eight putative species identified in previous molecular studies remain taxonomically unassessed (Supplementary Table S1).

To resolve the long-standing taxonomic instability within *Uropsilus*, we conducted extensive sampling across the MSC and adjacent ranges, generated genome-wide resequencing data from recognized and putative lineages, and conducted detailed traditional and geometric morphometric analyses. We applied this integrative framework to reassess species boundaries throughout the genus and provide formal taxonomic treatments, including updated diagnoses and a comprehensive identification key. By integrating coalescent-based species tree inference, species delimitation incorporating speciation process, gene flow detection, genome-wide measures of genetic divergence, cranial morphological and pelage color differentiation, and distribution range, we establish a rigorous, evidence-based framework for evaluating species limits in this morphologically conserved clade. This work aims to deliver the first stable taxonomic revision of *Uropsilus* and offers a generalizable model for species delimitation in other cryptic mammalian radiations.

2. Materials and methods

2.1. Sampling

We conducted whole-genome resequencing of 44 *Uropsilus* individuals representing all currently recognized species and most putative lineages distributed across the MSC and adjacent mountain systems (Fig. 1; Supplementary Table S2). For morphological analyses, we examined 253 museum specimens, including newly collected material covering the entire known geographic range of the genus (Supplementary Table S4).

Genomic sampling emphasized topotype or near-topotype localities for all described species to ensure accurate reference to nominal taxa. In addition, we included seven of the eight putative species proposed since Wan et al. (2013). The designations *U.* sp. 1–*U.* sp. 6 follow Bui et al. (2023), Ren et al. (2023), and Jiang et al. (2025). *U.* sp. 7 corresponds to "*U.* sp. 2" of Wan et al. (2013), and *U.* sp. 8 represents a newly identified lineage collected from Songpan County, Sichuan Province, during this study (Supplementary Table S1). Specimens of *U. fansipanensis* (described from two specimens from northern Vietnam) were unavailable for this study. Bui et al. (2023) reported that *U.* sp. 6 and *U. fansipanensis* form the closest sister relationship with the lowest divergence within the genus. We downloaded the *CYTB* sequence of the *U. fansipanensis* type specimen from NCBI (accession: OQ509373.1; OQ509374.1), which yielded a genetic distance of approximately 3.76% from *U.* sp. 6 — one of the lowest inter-lineage divergences recorded in the genus (Supplementary Table S3). Based on the key diagnostic characters provided in the original description of *U. fansipanensis*, including fur coloration and the relative size of premolars and canines, no discernible difference was observed, and no obvious geographic barrier between their distributions. Taking these lines of evidence together with nomenclatural considerations, we refer to *U.* sp. 6 as *U.* cf. *fansipanensis* to reflect that current data are insufficient to support their recognition as distinct species, while acknowledging that a definitive assessment requires direct specimen comparison. As *U.* sp. 6 (hereafter *U.* cf. *fansipanensis*) was retained as an independent analytical unit forming a terminal sister lineage to *U. fansipanensis*, this treatment has no bearing on the species delimitation or morphological comparisons of the remaining putative species.

For the morphological analyses in this study, we ensured that only adult individuals were included by confirming that all permanent molars (M^1–M^3 / M_1–M_3) were fully erupted. Furthermore, since no pronounced sexual dimorphism sufficient to confound interspecific comparisons has been reported in morphological studies of talpids and soricids, and pooling sexes has become a common practice in morphological and geometric morphometric analyses in these taxa including in *Uropsilus* (Loy et al., 1996; Rohlf et al., 1996; Tse and Calede 2021; Hu et al., 2021; Ren et al., 2023; Pillay et al., 2024; Jiang et al., 2025; Cao et al., 2026), we pooled males and females in all analyses. Species identification was based on an integrative assessment of (1) external morphological characters; (2) geographic information (locality and elevation), which was particularly informative given the allopatric or elevationally segregated distribution of most *Uropsilus* species; and (3) where available, *CYTB* molecular data, whereby specimens showing $\leq$ 2% sequence divergence from type specimens were treated as conspecific (Baker & Bradley, 2006).

All newly collected tissues were obtained under permits issued by the Animal Care and Ethics Committee of the Kunming Institute of Zoology (KIZ), Chinese Academy of Sciences, and by the relevant provincial wildlife authorities. Muscle or liver tissues were preserved in ethanol in the field and stored at –80 °C. All tissues and voucher specimens are deposited in the Kunming Natural History Museum of Zoology, KIZ, CAS. One additional skin-and-skull specimen (SAF-10130) is housed at the Sichuan Academy of Forestry (SAF).

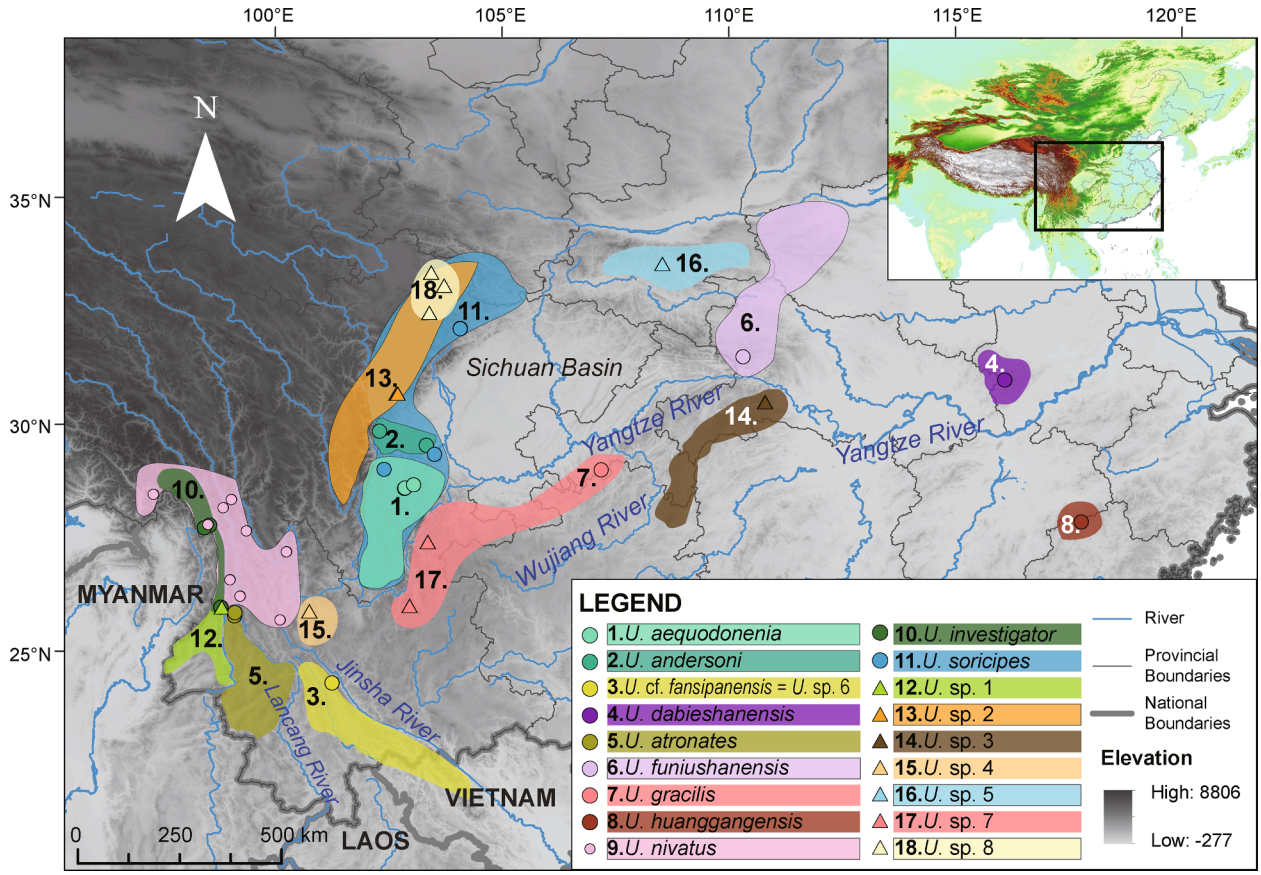


Fig. 1. Sampling locations for genome sequencing and species distribution ranges of *Uropsilus*. Different semi-transparent color polygons represent the estimated known distribution ranges of accepted and newly described species as of 2025, following the taxonomic assessment in this study.

2.2. Genomic resequencing and variant calling

Whole-genome resequencing was performed on the Illumina NovaSeq 6000 and DNBSEQ platforms (Supplementary Table S2). Raw sequencing reads were quality-controlled and preprocessed using Trimmomatic v0.39 (Bolger et al., 2014) to remove low-quality bases and adapter sequences. The resulting high-quality reads were mapped to the *Uropsilus gracilis* reference genome (Liu, Hua, et al., 2025) using BWA-MEM v0.7.17 (Li 2013).

Mapped reads (Binary Alignment/Map [BAM] files) were processed with Picard v3.1.1 to sort alignments and remove PCR duplicates. Single-nucleotide polymorphisms (SNPs) and genotypes were called using the Genome Analysis Toolkit (GATK v4.2.0.0) (McKenna et al., 2010) with stringent filtering criteria. All VCF files were subsequently merged, and SNPs with a missing data rate $< 20\%$ were retained using VCFtools v0.1.16 (Danecek et al., 2011). Detailed parameter settings for read processing, mapping, variant calling, and filtering are provided in the Supplementary Methods.

2.3. Species tree and divergence time

We used two coalescent-based approaches to infer the species tree. First, we applied SVDQuartets (Chifman and Kubatko 2014) to infer species tree topology from the SNPs dataset. High-quality biallelic SNPs obtained after quality control were thinned by retaining one SNP every 1,000 sites, resulting in 1,767,970 biallelic SNPs. One highly admixed individual (hybrids of *Uropsilus investigator* and *U.* sp. 1) was excluded from this analysis, as hybridization violates the assumptions of the multispecies coalescent model. We converted VCF files to NEXUS format and implemented the SVDQuartets function in PAUP* 4 (https://paup.phylosolutions.com/) to construct the species tree. *U. investigator* and *U.* sp. 1, which are the basal lineages within *Uropsilus* (Wan et al., 2018; He et al., 2021; Jiang et al., 2025; Li et al., 2026), were designated as the outgroup for rooting purpose. We performed two phylogenetic analyses: (1) an unconstrained reconstruction in which each individual was treated as a separate operational taxonomic unit (OTU) to test whether lineages formed exclusive clades, and (2) a constrained analysis in which the monophyly of each described or putative species was enforced.

Second, we inferred a species tree using ASTRAL v5.7.8 (Zhang et al., 2025) from genome-wide gene trees. For each individual, consensus sequences were generated from the sorted, duplicate-removed BAM files using ANGSD v0.935 (Korneliussen et al., 2014). The genome was partitioned into 10-kb windows with a 50-kb step size, and windows containing $> 20\%$ missing data were excluded. This filtering yielded 26,481 non-overlapping 10-kb alignments. We estimated maximum-likelihood gene trees for each alignment using IQ-TREE v2.4.0 (Minh et al., 2020), and then used ASTRAL to infer a species tree from the resulting collection of gene trees.

We also reconstructed a mitochondrial phylogenetic tree for all species and putative species. The mitochondrial genome was assembled from 5 Gb of clean reads per sample using GetOrganelle v1.7.2 (Jin et al., 2020) and annotated with MitoZ v2.3 (Meng et al., 2019). The *CYTB* sequences of two type specimens of *U. fansipanensis* downloaded from NCBI were incorporated into the dataset for combined analysis. The 13 protein-coding genes and 2 rRNA genes were subjected to multiple sequence alignment using ClustalW v2.1 (Larkin et al., 2007). The 15 aligned gene sequences were concatenated using AMAS (Borowiec 2016), which simultaneously generated the corresponding partition file. A maximum likelihood (ML) tree was reconstructed from the concatenated matrix using IQ-TREE v2.4.0 (Minh et al., 2020). The best-fit nucleotide substitution model for each gene partition was selected and partitions with similar models were merged to reduce parameter

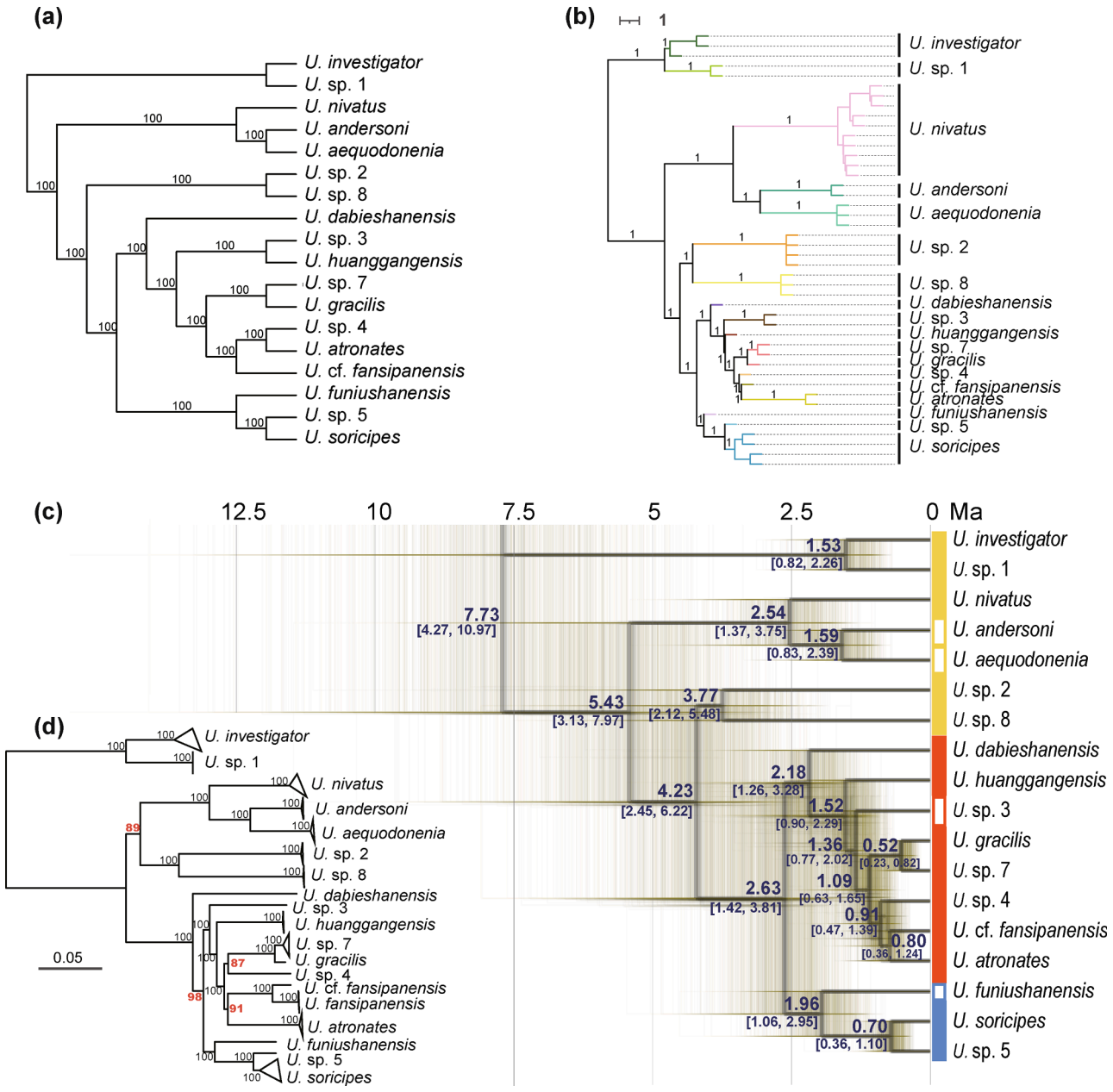


Fig. 2. Phylogenetic trees and divergence times in *Uropsilus*. (a) Species tree constructed by SVDQuartets based on SNPs data. (b) Species tree inferred by ASTRAL from 26,481 gene trees, with branch lengths in coalescent units. (c) DensiTree visualization of divergence times estimated by SNAPP. Each thin dark green line represents a posterior tree, and line density indicates uncertainty and consensus in divergence-time estimates; the thickest line represents the maximum clade credibility tree. Numbers above and below nodes indicate mean divergence times and 95% HPD (Highest Posterior Density) intervals, respectively. Colored vertical bars indicate the three species groups analyzed separately in the morphological PCA and LDA; hollow segments denote species not included in morphometric analyses (lineages with unique dentition or insufficient specimen numbers). (d) Maximum likelihood (ML) phylogenetic tree based on mitochondrial genomes. All lineages were constructed using 13 protein-coding genes and 2 rRNA genes from the mitochondrial genome, except for *U. fansipanensis*, which was represented only by the *CYTB* gene. Bootstrap support values below 100% are shown in red bold numbers. Branch lengths are in substitutions per site.

redundancy, implemented via the "-m MFP + MERGE" option. Node support was evaluated using ultrafast bootstrap approximation (UFBoot) with 1,000 replicates.

Divergence times among *Uropsilus* lineages were estimated using the SNAPP package (Bryant et al., 2012) in BEAST2 v2.6.6. To balance precision and computational efficiency, we retained 1,000 SNPs spaced at least 1,000 bp apart across the genome (Leache et al., 2014; Stange et al., 2018). The Markov chain Monte Carlo (MCMC) analysis was run for 1,000,000 iterations, with the first 20% discarded as burn-in. MCMC convergence and effective sample sizes (ESS) for all parameters were assessed using Tracer v1.7.2 (Rambaut et al., 2018), and ESS values > 200 were interpreted as evidence of adequate mixing and stationarity. A maximum clade credibility tree was summarized using TreeAnnotator v2.6.6 (Helfrich et al., 2018). Based on the estimated earliest divergence time of *Uropsilus* from He et al. (2021) at around 8.1 Ma, we set the calibration point for crown *Uropsilus* using a lognormal distribution (mean = 8.1, standard deviation = 0.22).

2.4. Species delimitation

We first quantified genomic differentiation between lineages and genetic diversity within lineages by computing pairwise absolute

nucleotide divergence (D_{XY}) and nucleotide diversity (π) in sliding windows using pixy v2.0.0 (Korunes and Samuk 2021). Both invariant sites and SNPs were included, and window size was set to 10,000 bp. To visualize the genome-wide distribution of genetic differentiation, we generated smoothed density plots of D_{XY} values across all windows. We also calculated pairwise inter-lineage Kimura 2-parameter (K2P) genetic distances based on *CYTB* sequences using MEGA v11.0.13 (Tamura et al., 2021), including the type specimens of *U. fansipanensis*.

For species delimitation, we used the DELINEATE framework (Sukumaran et al., 2021), which evaluates species boundaries relative to an existing taxonomic framework. Following the program authors' recommendations (https://jeetsukumaran.github.io/delineate/), we first subdivided samples as finely as possible by geographic population and used the "A11" module of BPP v4.8.2 (Flouri et al., 2018) to perform population delimitation and population tree estimation. From the set of non-adjacent windows used in the ASTRAL analysis, we randomly selected 2,000 windows (loci) and trimmed each to 1,000 bp for the BPP analysis. The MCMC analysis was run for 200,000 iterations after a burn-in of 30,000 iterations, sampling every 5 iterations. The posterior sample of trees was summarized using the sumtrees.py function from DendroPy v5.0.2 (Moreno et al. 2024) to generate a maximum clade credibility population tree. This ultrametric tree served as the guide tree for the subsequent DELINEATE analyses.

We then assigned species identities to each population delimited by BPP. First, we fixed the 11 currently accepted species as "known species" and tested species boundaries in the remaining seven lineages. For each candidate lineage, we summed the marginal probabilities (MPs) of the hypotheses that it represents a "new species", an "exclusive new species", or is part of an "existing species", thereby assessing the extent to which DELINEATE supports recognition of that lineage as a distinct species. For the following sister-lineage pairs, we additionally summed the MPs supporting their "conspecific" status to test whether each putative species has diverged sufficiently from its closest sister lineage to warrant species-level recognition: (*U. investigator* vs. *U.* sp. 1), (*U.* sp. 2 vs. *U.* sp. 8), (*U. huanggangensis* vs. *U.* sp. 3), (*U. gracilis* vs. *U.* sp. 7), (*U. atronates* vs. *U.* sp. 4) and (*U. soricipes* vs. *U.* sp. 5).

To further guard against taxonomic over-splitting, we conducted a supplementary analysis under a more conservative taxonomic framework that included only seven recognized species (*U. aequodonenia*, *U. andersoni*, *U. dabieshanensis*, *U. funiushanensis*, *U. investigator*, *U. nivatus*, and *U. soricipes*) as "known species". We used the sister-species pair *U. aequodonenia*–*U. andersoni*, which exhibits pronounced dental differences, as a benchmark and recognized only species whose divergence time and genetic distance exceeded those observed between these two species as reliably established "known species". Within this reduced framework, the remaining 11 lineages were treated as candidates. This conservative approach allowed us to test species boundaries while minimizing the risk of inflating species numbers.

2.5. Gene flow analyses

We assessed patterns of introgression among lineages using the f-branch statistic and two phylogenetic network inference approaches. First, we used the Dtrios module in Dsuite v0.4 r43 (Malinsky et al., 2021) to compute ABBA–BABA (*D*) statistics. We then applied the Fbranch function to summarize these results and to estimate the f-branch (f_b) statistic. Since no suitable outgroup was available for gene flow detection for all *Uropsilus* species, and the utility of the two phylogenetically basal lineages, *U. investigator* and *U.* sp. 1, as outgroups for detecting inter-lineage gene flow within this group has been validated (Li et al., 2026), we used these two lineages as outgroups to test for gene flow among the remaining 16 lineages.

To further investigate reticulate evolutionary relationships, we reconstructed phylogenetic networks using two methods: the maximum pseudolikelihood (MPL) approach implemented in PhyloNet (Yu and Nakhleh 2015) and the SNaQ method implemented in PhyloNetworks (Solís-Lemus and Ané 2016). We rooted the tree using *U. investigator* and *U.* sp. 1 for the 26,481 gene trees used for the ASTRAL analysis and input them for the MPL analysis in PhyloNet. For the SNaQ analysis, gene trees were first converted into quartet concordance factors (CFs), which were then used to infer networks. For both methods, we explored models allowing from 0 to 5 hybridization events (h = 0–5). Each analysis was run independently 10 times and the results of the best run (with the highest total log probability or lowest −loglik value) were retained.

2.6. Morphological analyses

We recorded external morphological data, including body weight (Wt), head–body length (HBL), tail length (TL), hind foot length (HF), and ear length (EL), from specimen labels or field notes. In addition, we measured 23 cranial characteristics using an electronic caliper with a precision of 0.01 mm (see Supplementary Methods). For all variables, we summarized ranges, means, and standard deviations. Cranial measurements were $\log_{10}$-transformed prior to analysis.

We conducted principal component analysis (PCA) and linear discriminant analysis (LDA) of cranial measurements in PAST v5.0.2 (Hammer et al., 2001). PCA was used to explore overall patterns of morphometric variation, while LDA was used to evaluate the discriminability of lineages identified in our phylogenomic analyses, with species/putative species assignments predefined accordingly. *U. andersoni* and *U. aequodonenia* were excluded from both analyses because they can be readily diagnosed by their genus-unique dental formula. *U. funiushanensis* and *U.* sp. 3 were also excluded due to insufficient sample sizes ($n < 3$). We first performed LDA on all 14 remaining lineages simultaneously, but neither analysis clearly separated most of them (Supplementary Fig. S3). Given that our primary aim is to test cranial discriminability among closely related lineages whose species boundaries are at issue, rather than among lineages already diagnosable by dental formula, pelage color, or tail proportions, we partitioned the 14 lineages into three groups based on phylogenetic relationships and shared external diagnostic characters, and conducted separate analyses within each group to improve resolution (Fig. 2c):

- **Group I** (*U. investigator*, *U.* sp. 1, *U. nivatus*, *U.* sp. 2, and *U.* sp. 8): lineages sharing dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38, with relatively long tails (mean TL/HBL > 0.90). These lineages exhibit ancient divergence and strong barriers to gene flow among them.
- **Group II** (*U. dabieshanensis*, *U. huanggangensis*, *U.* sp. 7, *U. gracilis*, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*): lineages also sharing dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38 but typically showing relatively shorter tails (mean TL/HBL < 0.90), except for *U. gracilis* (0.92), *U.* cf. *fansipanensis* (1.01), and *U. atronates* (0.96), which instead can be distinguished from Group I by their brighter reddish-brown pelage (versus dull brown or grey pelage in Group I). This group represents a rapidly radiating species complex characterized by significant incomplete lineage sorting (ILS) and gene flow, which has led to considerable taxonomic controversy.
- **Group III** (*U.* sp. 5 and *U. soricipes*): sister lineages sharing dental formula I 2/1, C 1/1, P 3/3, M 3/3 = 34, clearly distinguishing them from all other congeners.

This partitioning strategy facilitated visualization of interspecific morphological variation and enabled comparisons of morphological differentiation across lineages with contrasting divergence histories and evolutionary backgrounds. We applied Bartlett's test of sphericity and the Kaiser–Meyer–Olkin (KMO) measure to assess the suitability of the data for PCA in PAST. The PCA was performed on the correlation matrix, and missing data were handled using interactive imputation. Given the large number of variables — particularly the geometric morphometric landmark data — LDA was performed on the principal component scores

accounting for at least 95% of the total variance to reduce dimensionality and avoid overfitting. Lineages were used as the grouping variable to evaluate the discriminatory power of cranial measurements for classifying them. Classification accuracy of LDA was assessed using jackknifed (leave-one-out) cross-validation.

For geometric morphometric analyses, we photographed the dorsal and ventral views of the cranium and the lateral view of the mandible of each skull using a Sony ILCE-7RM5 digital camera, maintaining consistent orientation and distance between the camera and specimens. Images were converted to TPS format using tpsUtil v1.83, and landmarks were digitized with tpsDig v2.31 (Adams et al., 2004). A total of 10, 15, and 9 fixed landmarks were defined for the dorsal, ventral, and mandibular views of the cranium, respectively (see Supplementary Methods). For the mandible, we additionally placed 20 semi-landmarks at equal intervals along the contours of the inferior border of the horizontal ramus and the anterior border of the ascending ramus (Zelditch et al., 2004; Barrow and Macleod 2008). Only specimens without missing landmark data were retained for geometric morphometric analyses. Procrustes superimposition was first used to remove non-shape variation due to translation, rotation and scale. We then performed 2D landmark PCA in the "Geometry" module of PAST, followed by LDA with lineages as the grouping factor. Because ventral cranial shape variation generally provided the greatest power to distinguish between sister lineages, we further visualized ventral cranial shape differences among sister lineages involving candidate species using the "Thin-plate Spline" tools in PAST.

3. Results

3.1. Species tree reconstructions

We recovered molecular lineages corresponding to 17 previously recognized or putative species and one newly discovered lineage (*U.* sp. 8), an independent sister lineage to *U.* sp. 2, yielding a total of 18 lineages (Fig. 2a, b, c; Supplementary Fig. S1 and S2). The ASTRAL and BPP species trees showed identical topologies among these 18 lineages. The SNAPP divergence analysis was largely congruent with these trees, and differed only in one respect: it placed the divergence of *U.* sp. 3 earlier than that of *U. huanggangensis*. In contrast, SVDQuartets inferred *U.* sp. 3 and *U. huanggangensis* as sister lineages and supported the relationship ((*U.* cf. *fansipanensis*, (*U.* sp. 4, *U. atronates*))), unlike the ((*U.* sp. 4, (*U.* cf. *fansipanensis*, *U. atronates*))) supported by ASTRAL, BPP, and SNAPP. In the mitochondrial ML tree, in addition to the previously reported mito-nuclear discordance involving *U.* sp. 2 (represented in this study by the branch comprising *U.* sp. 2 + *U.* sp. 8) and *U. dabieshanensis* (Li et al., 2026), a topology of (*U.* sp. 4, (*U. gracilis*, *U.* sp. 7)) was also recovered (Fig. 2d). The branches exhibiting these discordant relationships were characterized by relatively low bootstrap support values and short internal branch lengths in the mitochondrial tree. The phylogenetic relationships of all remaining lineages were consistent with the species tree inferred by SNAPP and received 100% bootstrap support.

The SNAPP divergence-time estimates (Fig. 2c) showed that among the most recent divergences, the split between *U.* cf. *fansipanensis* and *U. atronates* occurred at approximately 0.80 Ma (95% highest posterior density [HPD]: 0.36–1.10 Ma). Two even more recent divergences involved putative species: *U. gracilis* vs. *U.* sp. 7 at around 0.52 Ma (95% HPD: 0.23–0.82 Ma) and *U. soricipes* vs. *U.* sp. 5 at around 0.70 Ma (95% HPD: 0.36–1.10 Ma). All remaining lineages diverged earlier than the *U.* cf. *fansipanensis*–*U. atronates* split. Notably, the sister lineages, *U.* sp. 2 and *U.* sp. 8, separated from all others as early as 4.23 Ma (95% HPD: 2.45–6.22 Ma) and then diverged around 3.77 Ma (95% HPD: 2.12–5.48 Ma).

3.2. Genomic species delimitation and genetic distances

Under the current taxonomic framework with 11 accepted species, the genomic delimitation framework recognized six of the seven additional lineages as new species, except for *U.* sp. 7 (Table 1). Specifically, the "new species" status of *U.* sp. 1, *U.* sp. 2, and *U.* sp. 8 received decisive support (MP > 0.95), whereas *U.* sp. 3, *U.* sp. 4, and *U.* sp. 5 received relatively strong support (MP = 0.78–0.86). In contrast, *U.* sp. 7 was not supported as a new species (MP = 0.05).

Tests of "conspecificity" indicated that the hypotheses that *U. investigator* vs. *U.* sp. 1, *U.* sp. 2 vs. *U.* sp. 8, *U. huanggangensis* vs. *U.* sp. 3, and *U. atronates* vs. *U.* sp. 4 are conspecific were all decisively rejected (MP ≤ 0.05). The pair *U. soricipes* vs. *U.* sp. 5 showed only weak support for being conspecific (MP = 0.17), whereas *U. gracilis* vs. *U.* sp. 7 received strong support for conspecificity (MP = 0.94).

When using the conservative taxonomic framework that included only seven recognized species (*U. aequodonenia*, *U. andersoni*, *U. dabieshanensis*, *U. funiushanensis*, *U. investigator*, *U. nivatus*, and *U. soricipes*), DELINEATE similarly supported the species boundaries of all candidate lineages except *U.* sp. 7 (Table 2). Overall support values were lower under this more conservative scenario, but the "new species" status of *U.* sp. 1, *U.* sp. 2, and *U.* sp. 8 remained strongly or decisively supported (MP > 0.90). The lineages *U. huanggangensis*, *U. gracilis*, *U.* cf. *fansipanensis*, *U. atronates*, *U.* sp. 3, *U.* sp. 4, and *U.* sp. 5 all received moderate to strong support (MP = 0.63–0.84). Consistent with the current framework analysis, *U.* sp. 7 was not supported as distinct and was instead decisively supported as conspecific with *U. gracilis* (MP = 0.96 for "conspecificity", Table 1).

D_{XY} calculated with a sliding-window approach across the genome ranged from 0.45% to 3.13% among the 18 lineages, and 0.50% to 3.02% among the currently recognized species (Fig. 3a). The smallest distance among the recognized species was observed between *U.* cf. *fansipanensis* and *U. atronates* (0.50%). Among pairs involving putative species, only the genetic distance between *U. gracilis* and *U.* sp. 7 (0.45%) fell below 0.50%. The D_{XY} distributions for pairs *U.* sp. 2 vs. *U.* sp. 8 and *U.* sp. 5 vs. *U. soricipes* are strongly right-skewed (Fig. 3b). In contrast, the D_{XY} distributions for pairs *U. investigator* vs. *U.* sp. 1, *U. gracilis* vs. *U.* sp. 7, and *U.* cf. *fansipanensis* vs *U.* sp. 4. are nearly symmetric and closely approximate a normal curve. Nucleotide diversity (π), which measures within-lineage genetic variation, varied markedly among lineages. In every case examined, π for a given lineage was lower than the D_{XY} value separating it from its closest sister lineage. It should be noted, however, that when a lineage is represented by a single individual, the π estimate reflects merely that individual's heterozygosity.

3.3. Inter-lineages gene flow patterns

We detected gene flow signals among multiple lineages within the recently diverged species complex (Group II), broadly consistent with findings from previous studies in *Uropsilus* (Li et al., 2026). The difference is that the inclusion of *U.* sp. 3 led to the detection of two additional

Table 1
Marginal probabilities (MP) of conspecificity from genomic species-delimitation analyses (inferred with DELINEATE). Lower MP values indicate stronger support for lineage-level separation.

Tested lineage pairs	MP of conspecificity		MP of exclusive conspecificity	
U. investigator & *U.* sp. 1	0.0203	*0.0767*	0.0203	*0.0767*
U. sp. 2 & *U.* sp. 8	0.0045	*0.0301*	0.0045	*0.0256*
U. huanggangensis & *U.* sp. 3	0.0272	*0.1388*	0.0215	*0.0152*
U. gracilis & *U.* sp. 7	0.9408	*0.9604*	0.8876	*0.5856*
U. atronates & *U.* sp. 4	0.0500	*0.2014*	0.0418	*0.0252*
U. cf. *fansipanensis* & *U. atronates*		*0.2078*		*0.0291*
U. soricipes & *U.* sp. 5	0.1729	*0.2902*	0.1690	*0.2412*

Note: Roman values report species delimitation results for the seven putative species under the current taxonomy, which includes 11 accepted species. Italic values report delimitation results under the conservative taxonomic framework, which recognizes only 7 species. (The same applies to Table 2.)

Table 2
Marginal probabilities (MP) of species status for putative new lineages.

Tested putative species	MP of new species		MP of existing species		MP of exclusive new species	
U. sp. 1	0.9697	*0.9133*	0.0203	*0.0767*	0.9697	*0.9133*
U. sp. 2	0.9885	*0.9727*	0.0015	*0.0173*	0.9835	*0.9358*
U. sp. 3	0.8615	*0.8358*	0.1286	*0.1543*	0.8413	*0.6465*
U. sp. 4	0.8125	*0.8348*	0.1775	*0.1552*	0.7928	*0.5754*
U. sp. 5	0.7798	*0.6287*	0.2102	*0.3613*	0.7748	*0.5940*
U. sp. 7	0.0463	*0.8350*	0.9438	*0.1550*	0.0451	*0.0242*
U. sp. 8	0.9885	*0.9727*	0.0015	*0.0173*	0.9835	*0.9358*
U. huanggangensis		*0.8356*		*0.1544*		*0.6423*
U. gracilis		*0.8350*		*0.1550*		*0.0242*
U. cf. *fansipanensis*		*0.8347*		*0.1553*		*0.5713*
U. atronates		*0.8347*		*0.1553*		*0.5713*

gene flow events (Fig. 4a): one between *U.* sp. 3 and the ancestral branch of (*U.* sp. 7, *U. gracilis*, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*), and another between *U. huanggangensis* and the ancestral branch of (*U.* sp. 3, *U.* sp. 7, *U. gracilis*, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*). Furthermore, unexpectedly strong gene flow was detected between *U.* sp. 8 and *U. soricipes*, *U.* sp. 5, and *U. funiushanensis* (Fig. 4a).

In the phylogenetic network analyses, the total log probability in PhyloNet plateaued when the number of reticulations (h) was increased from 0 to 1, and the − loglik in SNaQ stabilized at h = 2 (Fig. 4b, c). The optimal results of PhyloNet inferred a hybridization event between *U.* sp. 8 and *U. funiushanensis*, while the optimal results of of SNaQ inferred a hybridization event between *U.* sp. 8 and the ancestor of (*U. soricipes*, *U.* sp. 5 and *U. funiushanensis*) (Fig. 4b). SNaQ additionally recovered a hybridization signal involving *U. atronates* within the species complex (Fig. 4c), which was confirmed in previous studies through model testing as a case of distal ghost introgression (Li et al., 2026). The failure of PhyloNet to detect this signal may be attributable to a fundamental difference in computational approach: PhyloNet evaluates rooted triplets, whereas SNaQ employs quartet concordance factors. Taken together, both methods support a reticulate evolutionary signal involving *U.* sp. 8, which corresponds to the results of f-branch test.

3.4. Morphological differentiation

For Group I (*U. investigator*, *U.* sp. 1, *U. nivatus*, *U.* sp. 2, and *U.* sp. 8), PCA of linear cranial measurements showed that the first two principal components (PC1 and PC2) explained 70.87% of the total variance (Fig. 5). All loadings on PC1 were positive, and 13 of the 23 variables—primarily those related to overall length, width, and tooth row length—had loadings > 0.8, indicating that PC1 primarily represents overall skull size (Supplementary Table S5). In the PCA scatterplot, lineages were generally distinguishable along PC1 and PC2, with the sister pairs *U. investigator* vs. *U.* sp. 1 and *U.* sp. 2 vs. *U.* sp. 8 completely separated along PC1. In geometric morphometric analyses, some overlap occurred among lineages, but sister lineages could still be reliably distinguished. The LDA results showed that the lineages in this group, particularly sister lineages, could be basically distinguished based on traditional measurements and at least two sets of landmark data (Fig. 5). The overall classification accuracy after jackknifed cross-validation ranged from 79.59% to 90.24% (Supplementary Table S6). The misclassification rates between the two pairs of sister species were consistently low. *U.* sp. 8 had a relatively high misclassification rate, but misclassified specimens were mainly assigned to *U. nivatus* rather than its sister lineage *U.* sp. 2 (Supplementary Table S6).

For Group II (*U. dabieshanensis*, *U.* sp. 3, *U. huanggangensis*, *U.* sp. 7, *U. gracilis*, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*), PCA of any single dataset did not fully separate all lineages (Fig. 5). However, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates* could be differentiated using traditional cranial measurements and ventral cranial shape, whereas *U. dabieshanensis* and *U. huanggangensis* were separable based on both dorsal and ventral cranial shape. In contrast, *U.* sp. 7 and *U. gracilis* overlapped extensively in all datasets for PCA. The LDA based on traditional measurements yielded an overall classification accuracy of 83.46%, with misclassifications occurring mainly among *U.* sp. 7, *U. gracilis* and *U.* cf. *fansipanensis*, while the remaining lineages were well discriminated (Supplementary Table S6). The classification accuracy based on landmark data was considerably lower (51.61%–55.70%), indicating poor overall discrimination. Notably, the three terminal closely related lineages, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*, were effectively distinguished from one another in both traditional measurements and ventral cranial geometric morphometrics.

For Group III (*U.* sp. 5 and *U. soricipes*), PCA of traditional cranial measurements and mandible shape revealed clear separation between the two lineages (Fig. 5). The LDA based on traditional measurements and mandibular landmarks effectively discriminated *U.* sp. 5 and *U. soricipes*. In particular, the misclassification rate based on mandibular landmarks was zero, suggesting stable differentiation in mandible shape between the two lineages (Supplementary Table S6). Except for *U.* sp. 7 and *U. gracilis*, all sister lineages, including the closely related trio *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*, were clearly differentiated by linear craniometric data. Geometric morphometrics showed that ventral cranial shape was more pronouncedly differentiated among closely related lineages compared to the dorsal cranium and the mandible (Fig. 5). We therefore focused on ventral cranial shape and used thin-plate spline visualizations to compare candidate species with their closest sister species (Supplementary Fig. S4). There was no significant shape deformation between *U.* sp. 7 and *U. gracilis*. *U. investigator* and *U.* sp. 1 were only slightly different based on overall shape, with the primary difference being the much smaller body size and cranial dimensions of *U.* sp. 1 relative to *U. investigator*; after removing size effects, differences did not deviate from those already captured by traditional measurements.

Other candidate species showed more substantial shape differences from their sister lineages. *U.* sp. 8 differed from *U.* sp. 2 mainly by having a shorter anterior rostrum and a pterygoid process extending further toward the midline. From *U. atronates* to *U.* cf. *fansipanensis* to *U.* sp. 4, there was a trend toward a narrower braincase and longer, broader upper molars. *U.* sp. 5 differed from *U. soricipes* in having a broader posterior rostrum and a shorter, narrower anterior rostrum. Because *U.* sp. 2 vs. *U.* sp. 8 and *U. soricipes* vs. *U.* sp. 5 could be clearly distinguished in PCA of mandibular landmarks, we generated additional TPS deformation plots for these pairs (Supplementary Fig. S5). Relative to *U.* sp. 2, *U.* sp. 8 showed a more slender horizontal and ascending ramus. Compared to *U. soricipes*, *U.* sp. 5 exhibited a more anteriorly inclined anterior margin of the ascending ramus (smaller angle relative to the mandibular body) and relatively longer lower molar rows.

Taken together, the cranial and mandibular morphometric analyses corroborate the genomic species delimitation results for all candidate species except *U.* sp. 7, which is both genomically and morphologically indistinguishable from *U. gracilis* and is therefore best treated as conspecific.

4. Discussion

Drawing on genome-wide data and multidimensional analytical methods, we systematically re-evaluated the taxonomic status of all known *Uropsilus* lineages. Our study represents the first comprehensive taxonomic revision of the genus, integrating genomic species delimitation, divergence-time estimation, and both traditional and geometric morphometrics. This endeavor is not only foundational for taxonomy but also critical for understanding speciation mechanisms and for guiding effective conservation.

4.1. Genomic species trees and species delimitation

Previous phylogenetic studies, which were based on larger sample sizes of genetic data, have convincingly demonstrated the monophyly of

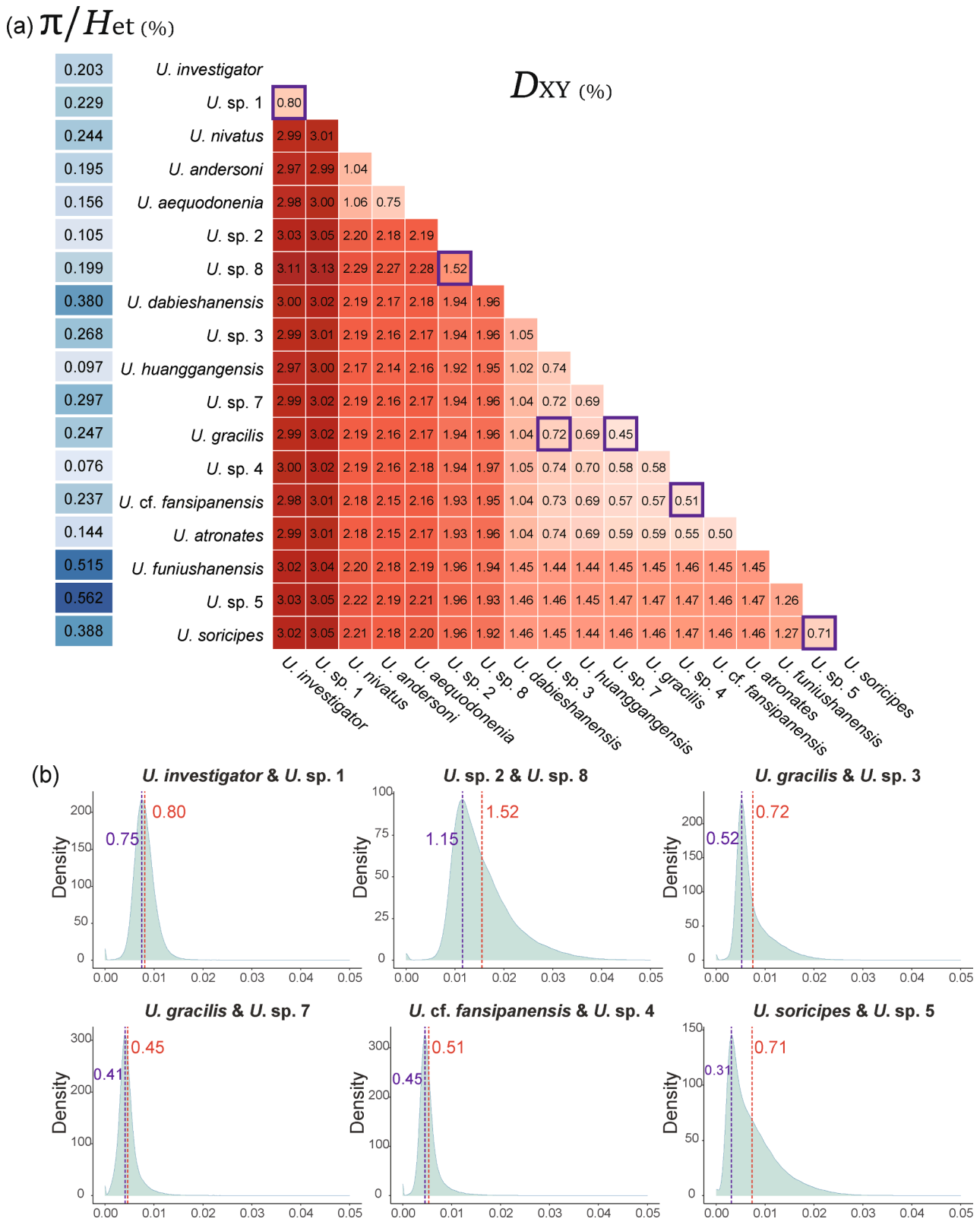


Fig. 3. Nucleotide diversity (π) within lineages and pairwise absolute genetic divergence (D_{XY}) among lineages. (a) Blue single-column heatmap showing π values for each lineage; for lineages represented by a single sample, π is estimated as heterozygosity (Het). The red heatmap shows pairwise D_{XY} between lineages. D_{XY} values between each putative species and its closest sister lineage are highlighted with deep purple boxes and correspond to panel (b). (b) Smoothed density distributions of genome-wide window-based D_{XY} for each focal pair. The red dashed line indicates the mean D_{XY}, and the blue dashed line indicates the mode (peak) estimated by kernel density estimation.

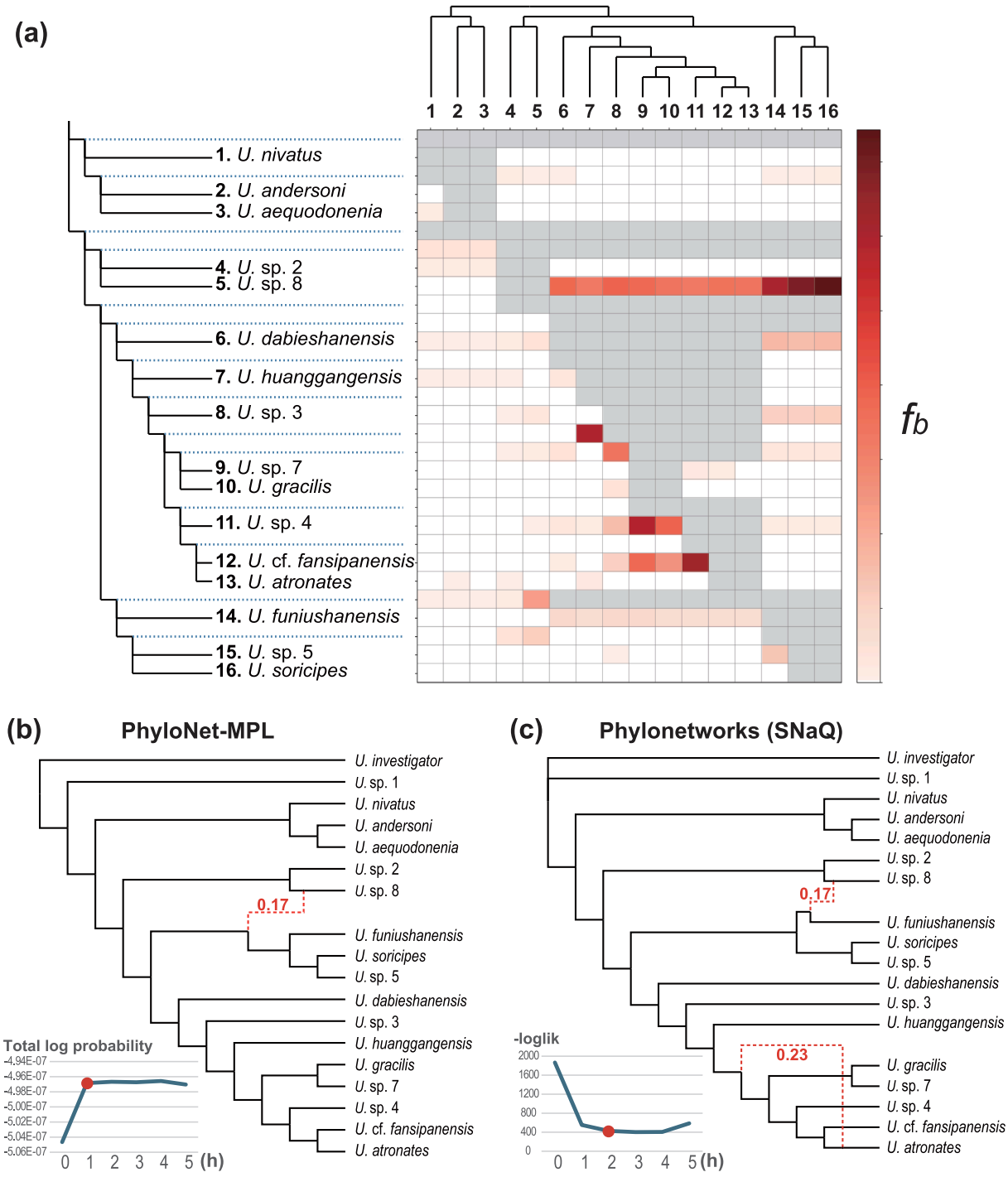


Fig. 4. Gene flow and phylogenetic network. (a) Results of f-branch analysis using Dsuite. (b, c) Best results of phylogenetic networks inferred by two methods. Minor reticulation edges and their inheritance probabilities are shown as red dashed lines with corresponding values. The bottom-left insets shows the comparison of results across five runs, with the red dot indicating the optimal result that reached a plateau.

each lineage, although conflicts in the phylogenetic relationships remain (Wan et al., 2018; Hu et al., 2021; Bui et al., 2023; Ren et al., 2023; Jiang et al., 2025). In this study, we applied genome-scale data and various analytical methods to focus on the construction of species trees, genomic differentiation, and species delimitation. The application of genomic data has greatly improved the resolution and stability of the inferred species trees. The topologies inferred by multiple approaches were generally consistent, with only minor discrepancies for the position of *U.* sp. 3 and *U.* sp. 4. Past phylogenetic conflicts were mainly concentrated among *U. dabieshanensis*, *U. huanggangensis*, *U. gracilis*, *U. fansipanensis*, *U. atronates*, *U.* sp. 3, *U.* sp. 4 and *U.* cf. *fansipanensis*, where extremely short internode branch lengths suggest a rapid radiation history resulting in strong incomplete lineage sorting (ILS) (Li et al., 2026). The gene flow analyses incorporating the newly included *U.* sp. 3 revealed that the phylogenetic uncertainty surrounding the placement of *U.* sp. 3 may be influenced by strong gene flow signals. Notably, both *U.* sp. 3 and *U. huanggangensis* were inferred to have experienced gene flow with the ancestral branches of multiple other lineages within the complex. This finding is consistent with a diversification scenario in which widespread dispersal and rapid radiation-like diversification occurred during the

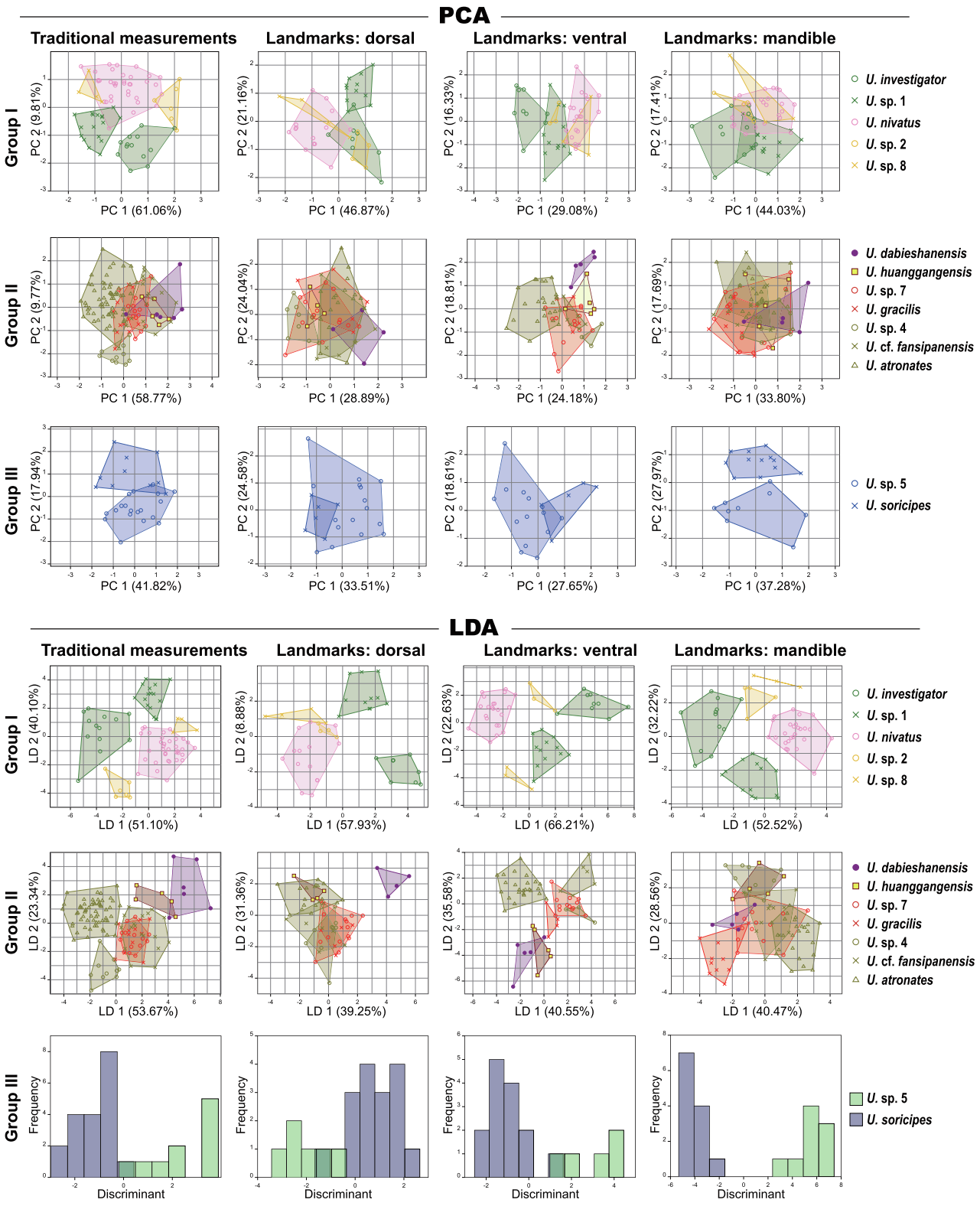


Fig. 5. Principal component analysis (PCA) and linear discriminant analysis (LDA) of morphological data from three species groups. Scatterplots show the first two principal components (PC1 and PC2) or linear discrimimant axes (LD1 and LD2) for: measurements of 23 traditional cranial traits; 10 dorsal cranial landmarks; 15 ventral cranial landmarks; and 9 landmarks with 20 semi-landmarks of mandible.

early Pleistocene, followed by prolonged geographic isolation that allowed fixed genomic differences to continue accumulating (Wan et al., 2019; Li et al., 2026). Building upon previous studies, we further highlight the remarkable lineage diversity and explosive diversification history of this species radiation within the genus. Meanwhile, our findings underscore the value of genome-scale data and coalescent-based species tree methods in providing critical insights into such complex evolutionary histories.

The DELINEATE delimitation framework incorporates a birth–death model and conditions on the existing taxonomic framework, effectively

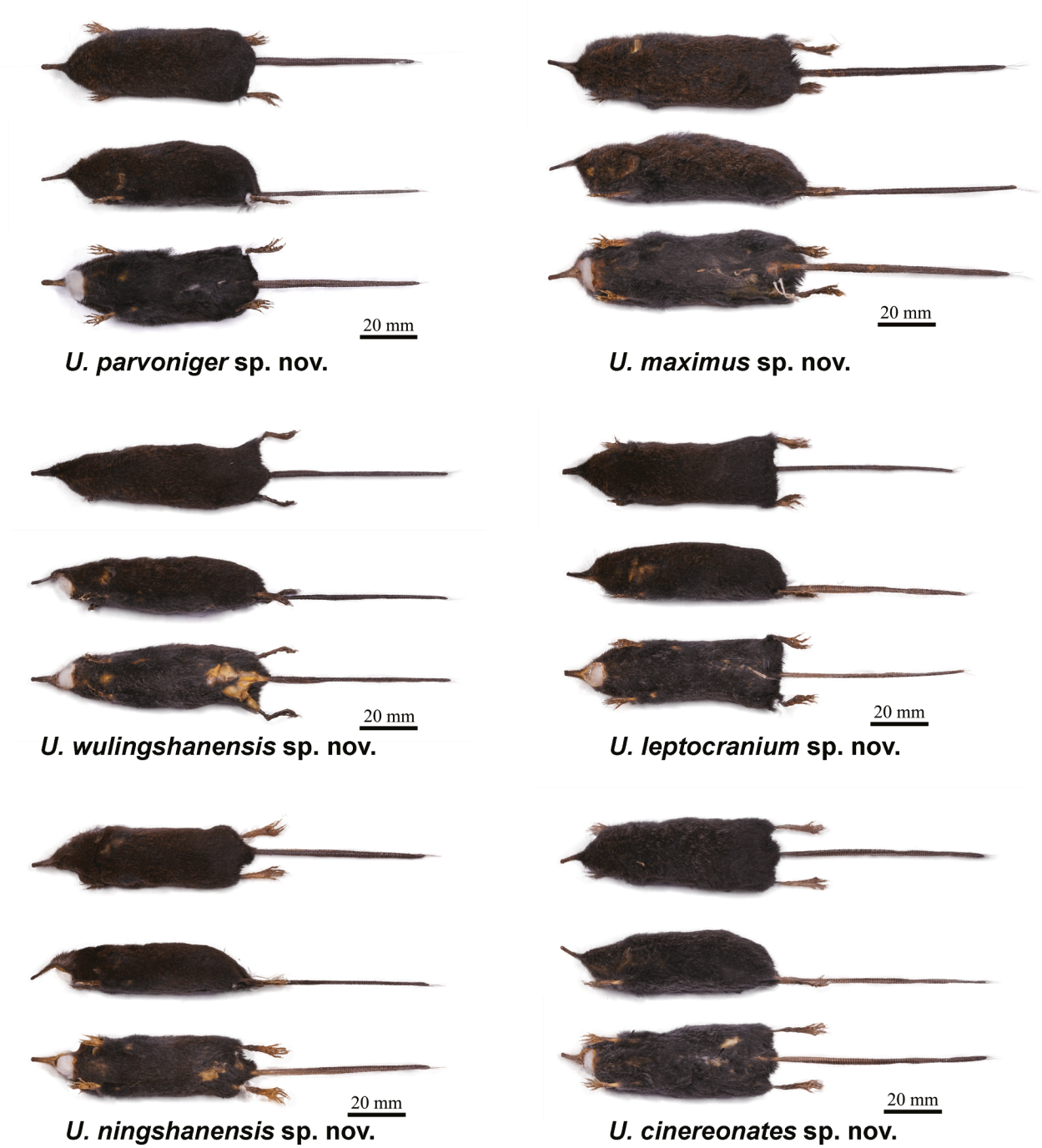


Fig. 6. Holotypes skins of six new species. Dorsal, ventral, and lateral views of the holotypes skins of *Uropsilus parvoniger* sp. nov. (KIZ-042715), *Uropsilus maximus* sp. nov. (KIZ-028549), *U. wulingshanensis* sp. nov. (KIZ-042083), *U. leptocranium* sp. nov. (KIZ-022457), *U. ningshanensis* sp. nov. (KIZ-016111) and *Uropsilus cinereonates* sp. nov. (KIZ-042984).

accounting for false inflations of population-level structure as species-level divergences, particularly in multispecies coalescent (MSC) analyses (Sukumaran et al., 2021; Karbstein et al., 2024). The tendency of genomic data to yield highly resolved relationships while being susceptible to oversplitting lineages has been highlighted for BPP and Bayes factor delimitation (BFD) analyses (Leache et al., 2014; Sukumaran and Knowles 2017; French et al., 2019; Ferrer Obiol et al., 2023). In our case, BPP treated all geographically defined populations as distinct species with extremely high posterior probabilities, illustrating this risk. Under both the current 11-species taxonomy and a more conservative seven-species framework, DELINEATE consistently supported six of the seven putative lineages as independent species, and *U. sp.* 7 as conspecific with *U. gracilis*. These results agree with similar empirical applications of the DELINEATE framework in other systems, where the method has proven helpful in discovering and evaluating new species while anchoring the prevailing taxonomy (Calixto-Rojas et al., 2023; Chan et al., 2023). However, the relatively lower MP values for *U.* sp. 3, *U.* sp. 4, and *U.* sp. 5 (0.78–0.86) reflect the recent divergence of these lineages, whose accumulated genetic differentiation may not yet have reached levels typical of well-established species within the genus, and thus introduce non-negligible uncertainty into their species-level recognition. More broadly, MSC-based approaches do not explicitly model post-divergence gene flow or hybridization (Flouri et al., 2018, 2019), and limited sample sizes for certain lineages may bias parameter

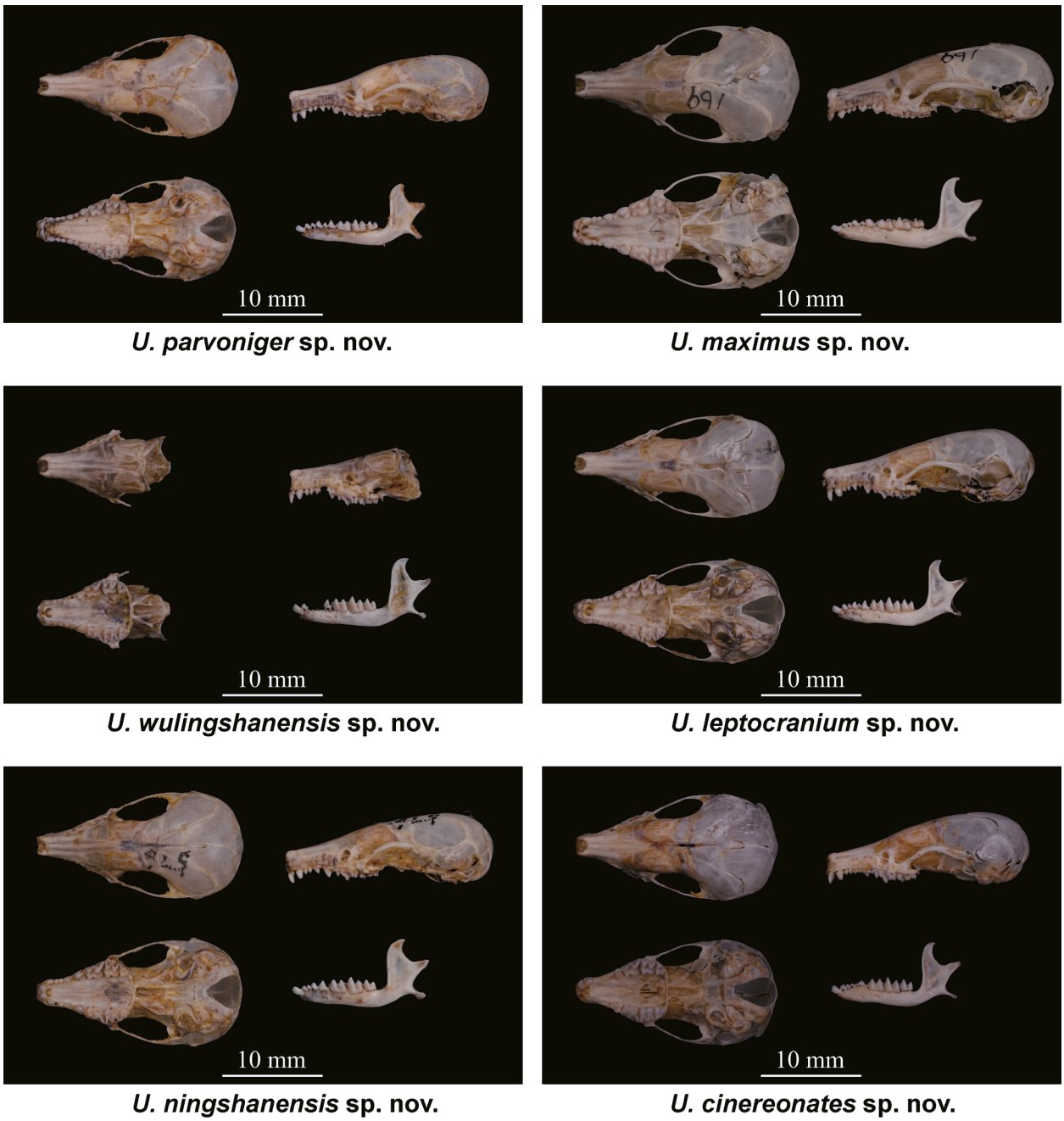


Fig. 7. Holotypes skulls of six new species. Dorsal, ventral, and lateral views of the craniums and lateral views of the mandibles of the holotypes of *Uropsilus parvoniger* sp. nov. (KIZ-042715), *Uropsilus maximus* sp. nov. (KIZ-028549), *U. wulingshanensis* sp. nov. (KIZ-042083), *U. leptocranium* sp. nov. (KIZ-022457), *U. ningshanensis* sp. nov. (KIZ-016111) and *Uropsilus cinereonates* sp. nov. (KIZ-042984).

estimates and species delimitations. Under the conservative framework, boundaries between some closely related lineages became more ambiguous, highlighting how prior taxonomic decisions can influence downstream inference. These limitations collectively underscore the necessity of integrating multiple independent lines of evidence within a well-established taxonomic framework to achieve robust species delimitation.

Over the last decade, recognition of new or putative *Uropsilus* species relied predominantly on a phylogenetic species concept (Zachos 2015), often with minimal morphological support. This reflected in small morphological sample sizes, a lack of comparative analyses among closely related lineages, and the absence of clear diagnostic keys. Such practices risk both over- and under-recognition of species and complicate subsequent ecological and evolutionary work. Our study addresses these shortcomings by jointly assessing absolute genomic differentiation (D_{XY}), divergence times, and quantitative morphological divergence, providing an integrative basis for taxonomic decisions.

4.2. Integrative assessment of putative species in Uropsilus

Based on the DELINEATE results above, we evaluated each putative species using absolute genomic divergence (D_{XY}), divergence times, and traditional and geometric morphometrics.

***U.* sp. 1.** *U.* sp. 1 was strongly supported as a distinct species by DELINEATE (MP = 0.97), with a 0.80% D_{XY} to *U. investigator*. Morphologically, *U.* sp. 1 has a markedly smaller body size and cranial

dimensions and a distinctly darker rump pelage (Table S7). The two lineages occur mainly in the northern–central (*U. investigator*) and central–southern (*U.* sp. 1) Gaoligong Mountains, with only limited sympatry and evidence for a few hybrids (Li et al., 2026; Fig. 1). Their divergence, at ~ 1.53 Ma, is older than several recognized species splits in the genus, and their genome-wide divergence exceeds that among all other sister-species pairs evaluated (Fig. 3). The presence of hybrids does not erode their overall genomic and morphological distinctness. We therefore recognize *U.* sp. 1 as a valid species.

***U.* sp. 2 and *U.* sp. 8.** The boundary between *U.* sp. 2 and *U.* sp. 8 was well supported (MP = 0.99), with D_{XY} of 1.52% and a divergence time of ~ 3.77 Ma. External pelage, traditional cranial measurements, and geometric morphometric landmarks distinctly differentiate these lineages. Their distributions partially overlap in the western Sichuan mountains (Fig. 1), but existing sampling suggests elevational separation. *U.* sp. 8 occurs above ~ 3,200 m and *U.* sp. 2 between 2,200–3,100 m, consistent with potential niche differentiation. Notably, strong gene flow signals were detected between *U.* sp. 8 and *U. soricipes*, *U.* sp. 5, and *U. funiushanensis*. However, the latter three species are all distributed below 2,500 m in elevation. Although the geographic range of *U. soricipes* overlaps horizontally with that of *U.* sp. 8, the considerable divergence time, pronounced morphological differences, and ecological niche differentiation collectively suggest that these three species have most likely established complete reproductive isolation from *U.* sp. 8. Combined with the results of phylogenetic network analyses, these gene flow signals likely reflect ancient hybridization events or ghost introgression. These species are all distributed around the Sichuan Basin, and previous studies have proposed that the mountains encircling the basin may have historically served as stepping stones facilitating dispersal and genetic exchange among populations of small mammals with limited dispersal ability, such as long-tailed moles and Asian shrew moles (He et al., 2019; Li et al., 2026). The detection of these gene flow signals actually lends further support to a pattern of overall landscape connectivity and sky island-driven speciation. Given the deep divergence, strong genomic differentiation, and consistent morphological differences, we elevate *U.* sp. 2 to species rank and confirm the species status of *U.* sp. 8.

***U.* sp. 3.** *U.* sp. 3 was strongly supported as a distinct species (MP = 0.86), with a mean D_{XY} of 0.72% from *U. huanggangensis*, *U. gracilis*, *U.* cf. *fansipanensis*, *U. atronates*, and *U.* sp. 7, and a divergence time of ~ 1.36 Ma. This divergence predates that among *U. gracilis*, *U.* cf. *fansipanensis*, *U. atronates*, and *U.* sp. 7 (~1.09–0.52 Ma). In *U.* sp. 3, the third upper premolar (P^3) is clearly smaller than the upper canine (versus equal to or larger in *U. huanggangensis*) (Supplementary Fig. S6), and *U.* sp. 3 has a shorter tail and darker, more uniform dorsal coloration than *U. gracilis*, *U.* cf. *fansipanensis*, *U. atronates*, and *U.* sp. 7 (Supplementary Table S7; Supplementary Fig. S7). Its known range is allopatric, separated from *U. gracilis*, *U. dabieshanensis*, and *U. funiushanensis* by major river barriers (Wujiang River and Yangtze River) (Fig. 1). The gene flow signals detected between *U.* sp. 3 and the ancestral branch of (*U.* sp. 7, *U. gracilis*, *U.* sp. 4, *U.* cf. *fansipanensis*, and *U. atronates*) (Fig. 4a), combined with the extremely short internal branch lengths and its unstable phylogenetic placement in both nuclear and mitochondrial genomic phylogenetic trees (Fig. 2), suggest that *U.* sp. 3 originated during the radiation of this species complex. In the context of long-term strict geographic isolation, the accumulation of extensive genomic differentiation (Fig. 3) supports the recognition of *U.* sp. 3 as a distinct species, although additional specimens would allow a more complete quantitative characterization.

***U.* sp. 7.** Unlike other lineages, *U. sp.* 7 was not supported as a distinct species (MP for "new species" = 0.05; MP for conspecificity with *U. gracilis* = 0.94). The D_{XY} between *U.* sp. 7 and *U. gracilis* was 0.45%, and their divergence time (~0.52 Ma) was the youngest among all lineage pairs. The K2P distance between them (1.35%) is notably lower than the interspecific genetic distances reported among known species within the genus (Supplementary Table S3). No stable differences were detected in external morphology, traditional cranial measurements, or geometric morphometrics. Although the two populations are geographically distant, they are not separated by obvious barriers (Fig. 1). *U.* sp. 7 was initially designated as *U.* sp. 2 by Wan et al. (2013), but subsequent studies sometimes merged it with *U. gracilis* (Table S1). Our integrative evidence supports this synonymy: *U.* sp. 7 is best treated as conspecific with *U. gracilis*.

***U.* sp. 4.** *U.* sp. 4 was supported as a distinct species (MP = 0.81), with D_{XY} to *U.* cf. *fansipanensis* and *U. atronates* of 0.51–0.55% and a divergence time of ~ 0.91 Ma, predating the split between *U.* cf. *fansipanensis* and *U. atronates* (~0.80 Ma). While these three lineages are superficially similar in external appearance, cranial shape analyses reveal a clinal pattern from *U. atronates* to *U.* cf. *fansipanensis* to *U.* sp. 4, and PCA/LDA distinguish all three (Fig. 5). Their ranges do not overlap: *U.* sp. 4 and *U.* cf. *fansipanensis* occur east of *U. atronates*, which is separated from them by the Lancang River. The barrier between *U.* sp. 4 and *U.* cf. *fansipanensis* is less clear, potentially allowing limited ongoing gene flow. The diversification process among these closely related lineages may still have been accompanied by gene flow, thereby blurring their genetic species boundaries to some extent. Nevertheless, morphological analyses support significant differentiation among the three closely related species. In particular, given that *U. fansipanensis* has already been accepted as a valid species and that the divergence of *U.* sp. 4 predates that of *U. fansipanensis* (Bui et al., 2023) (Fig. 2; Fig. 3; Supplementary Table S3), we elevate *U.* sp. 4 to the status of a valid species.

***U.* sp. 5.** *U.* sp. 5 received moderate to strong support as a distinct species (MP = 0.78), with a mean D_{XY} to *U. soricipes* of 0.71% and a divergence time of ~ 0.70 Ma. The most pronounced differences lie in the mandible: *U.* sp. 5 has a smaller angle between the mandibular body and the anterior edge of the coronoid process, and a longer lower molar row (Supplementary Fig. S5). Such mandibular variation has been linked to dietary specialization in talpids (Barrow and Macleod 2008; He et al., 2013), suggesting possible ecological divergence. The D_{XY} distribution is strongly right-skewed, consistent with heterogeneous genomic divergence. This pattern could reflect recent divergence accompanied by strong selection on particular genomic regions ("speciation islands"), or older divergence partially obscured by recent gene flow. Geographically, *U. soricipes* occupies mid-elevation zones of the Sichuan mountains east of the Hengduan Mountains, whereas *U.* sp. 5 is restricted to the Qinling Mountains at the boundary between northern and southern China. Their genome-wide divergence exceeds that between recognized species such as *U. huanggangensis* vs. *U. gracilis* or *U.* cf. *fansipanensis* vs. *U. atronates*, and they also differ consistently in functionally relevant mandibular traits. We therefore recognize *U.* sp. 5 as a valid species.

In summary, integrating delimitation results, genomic divergence, divergence-time estimates, morphology, and geography, we elevate *U.* sp. 1, *U.* sp. 2, *U.* sp. 3, *U.* sp. 4, *U.* sp. 5, and *U.* sp. 8 to full species status, and provide formal names, diagnoses, and an updated identification key in section 4.4 and 4.5.

4.3. Evolutionary and taxonomic significance of fine-scale species diversity assessment

Uropsilus has long been regarded as a genus with high cryptic diversity, shaped by "sky island" montane habitats (He and Jiang 2014; Wan et al., 2018). Many closely related *Uropsilus* lineages have well-defined genomic boundaries and restricted gene flow (Li et al., 2026), with lineages previously considered cryptic exhibiting stable morphological divergence when examined with sufficient sample sizes and quantitative methods. In some cases, morphological differences between sister lineages exceeded those between more distantly related species.

The patterns of D_{XY} and π among sister lineages pointed to heterogeneous histories of divergence. For example, the pairs *U. investigator*–*U.* sp. 1 and *U. gracilis*–*U.* sp. 7 shared relatively homogeneous D_{XY} distributions and similar π values, consistent with largely neutral divergence

driven by geographic isolation (Fig. 3). However, the divergence between *U. investigator* and *U.* sp. 1 was older, and the two showed clear species boundaries, whereas *U. gracilis* and *U.* sp. 7, which remain conspecific, showed less divergence. In contrast, pairs such as *U.* sp. 2–*U.* sp. 8 and *U. soricipes*–*U.* sp. 5 exhibited strongly skewed D_{XY} distributions and marked differences in π, suggesting heterogeneous genomic divergence shaped by selection or complex demographic histories. These pairs also showed pronounced mandibular differentiation (Supplementary Fig. S5), consistent with potential dietary or functional divergence (Barrow and Macleod 2008; He et al., 2013). Together, these observations highlight that even within a morphologically conservative genus, species divergence can proceed along multiple trajectories—ranging from predominantly neutral isolation to more complex, possibly adaptive scenarios—and that robust species delimitations require integrating genomic, morphological, and geographic data.

The ongoing discovery of new *Uropsilus* species and range extensions (He et al., 2016; Wan et al., 2018; Li et al., 2026) suggests that species diversity in this genus remains potentially underestimated. This pattern mirrors a broader trend across small mammals in the montane regions of southwestern China, where numerous new genera and species have been described in recent years (Zhang et al., 2022; Chen et al., 2023; Wang et al., 2024; Liu, Wang, et al., 2025; Zhu et al., 2025). Establishing stable taxonomic systems and clear morphological identification criteria is therefore critical for *Uropsilus* and other small mammal clades in this region.

Given the complex evolutionary history of *Uropsilus* and the potential for sporadic hybridization to obscure external characters, future work should combine genome-wide data with broad geographic and ecological sampling. Such integrative efforts will help link genomic signatures of divergence with robust, field-applicable morphological characters and further refine species boundaries.

4.4. Taxonomic Account

Based on integrated phylogenomic analyses, coalescent species delimitation, absolute genetic distances, and both traditional and geometric morphometrics, our study supports the recognition of six previously unnamed lineages as valid species: *U.* sp. 1, *U.* sp. 2, *U.* sp. 3, *U.* sp. 4, *U.* sp. 5, and *U.* sp. 8. We formally name and describe these six new species as follows:

- *U.* sp. 1 = *Uropsilus parvoniger* sp. nov.
- *U.* sp. 2 = *Uropsilus maximus* sp. nov.
- *U.* sp. 3 = *Uropsilus wulingshanensis* sp. nov.
- *U.* sp. 4 = *Uropsilus leptocranium* sp. nov.
- *U.* sp. 5 = *Uropsilus ningshanensis* sp. nov.
- *U.* sp. 8 = *Uropsilus cinereonates* sp. nov.

4.4.1. *Uropsilus parvoniger Li YX, Li Q et Jiang XL, sp. nov. (Figs. 1, 6, 7)*

Zoobank LSID: urn:lsid:zoobank.org:act:30DBA726-5949-45C9-AEB7-9A053C1D0F56.

Holotype. KIZ-042715 (field ID: PM1312467), an adult male collected by Changzhe Pu on 2 December 2013 from Pianma Village, Lushui County, western Yunnan Province, China (25.978°N, 98.672°E; elevation 3,058 m a.s.l.). The dried skin, cleaned skull, and alcohol-preserved carcass deposited in the Kunming Natural History Museum of Zoology, Kunming Institute of Zoology (KIZ).

Paratypes. KIZ-042717 (field ID: PM1312422), adult female collected by Quan Li on 30 November 2013 from Pianma Village, Lushui County, western Yunnan Province, China (25.981°N, 98.674°E; 2,823 m a.s.l.); KIZ-042703 (field ID: PM1312570), adult female collected by Changzhe Pu on 4 December 2013 from Pianma Village, Lushui County, western Yunnan Province, China (25.977°N, 98.674°E; 2,997 m a.s.l.). The dried skins, cleaned skulls, and alcohol-preserved carcasses were deposited in the Kunming Natural History Museum of Zoology, KIZ.

Measurements. See Supplementary Table S5.

Diagnosis. A small-bodied species of *Uropsilus*. Among previously described congeners, only slightly larger than *U. atronates* but distinguished by darker, duller pelage. Externally similar to *U. investigator* but markedly smaller in body and skull size. Head and dorsum dark brown to black; ventral pelage dark grey to black, without a distinct boundary between dorsal and ventral coloration. Rump black, darker than head and dorsum. Tail uniformly dark dorsally and ventrally. Dental formula: I 2/1, C 1/1, P 4/4, M 3/3 = 38.

Description. Small-sized Asian shrew mole with body mass (Wt) 7.65 ± 1.00 g (range 5.9–9.5 g) and head–body length (HBL) 69.7 ± 2.82 mm (range 65–76.5 mm). Ears are relatively small (EL = 7.91 ± 1.11 mm, range 6.5–10 mm), but pinnae project above the pelage. Eyes minute and concealed by fur. Tail length/head–body length ratio (TL/HBL) 0.91 ± 0.09 (range 0.77–1.11); tail tip bearing several bristle hairs approximately 5 mm long. Dorsal pelage dark brown, grading into jet black on the rump. Dorsal hair tips brown, with dark grey to black bases. Ventral pelage matches dorsal hair bases (dark grey to black). No sharp boundary between dorsal and ventral coloration. Tail generally uniformly black dorsally and ventrally.

Skull small: profile length (PL) 20.49 ± 0.32 mm (range 19.97–21.08 mm), greatest neurocranium breadth (GNB) 10.86 ± 0.26 mm (range 10.37–11.28 mm), and mandible length (ML) 13.31 ± 0.20 mm (range 12.85–13.72 mm). Zygomatic arches not markedly expanded laterally (do not exceed greatest breadth of braincase). Infraorbital foramen anterior and ventral to lacrimal foramen and larger than the latter. Temporal crests separated by approximately one third of the least interorbital breadth. In lateral view, braincase shows an anterior bulge projecting beyond the line connecting nasals and occiput. Anterior margin of mandibular coronoid process forms an angle of ~ 90–100° with the mandibular body. Dentition: I 2/1, C 1/1, P 4/4, M 3/3 = 38. Upper canine larger than P^1 but smaller than P^2. P^3 the smallest upper premolar, equal to or smaller than P^1; P^4 well developed. In the mandible, only one incisor (I_2) is present and is enlarged and chisel-shaped. P_1 and P_3 equal in size and smaller than the lower canine. Lower canine, P_2, and P_4 increase in size sequentially.

Etymology. The specific epithet *parvoniger* is derived from the Latin *parvus* (small) and *niger* (black), referring to the combination of notably small body size and distinctive black coloration, especially on the rump and tail. The epithet is treated as an adjective and agrees in gender with the masculine genus *Uropsilus*.

Comparison. See Supplementary Material.

Distribution and ecology. *Uropsilus parvoniger* sp. nov. is known from the southern section of the Gaoligong Mountains, Yunnan Province, southwestern China, specifically Lushui and Tengchong counties. It inhabits bamboo, broadleaf, fir, and rhododendron forests at elevations of 2,600–3,500 m. The range likely extends westward into adjacent montane regions of Myanmar. *U. parvoniger* sp. nov. and *U. investigator* show limited sympatry in Lushui County, where occasional hybridization may occur (Li et al. 2025). Despite this, genomic divergence and cranial and external morphological differences between the two species are pronounced.

Remarks. *U. parvoniger* sp. nov. is most closely related to *U. investigator*, from which it can be distinguished by its notably smaller body and skull size and darker rump pelage relative to the dorsum. For *U. parvoniger* sp. nov.: PL = 19.97–21.08 mm, ML = 12.85–13.72 mm; for *U. investigator*: PL = 21.18–22.26 mm, ML = 13.91–14.81 mm.

4.4.2. *Uropsilus maximus Li YX et Jiang XL, sp. nov. (Figs. 1, 6, 7)*

Zoobank LSID: urn:lsid:zoobank.org:act:779A2542-0E12-40B2-AD47-4B888105F896.

Holotype. KIZ-028549 (field ID: 0905169), an adult male collected by Xuelong Jiang on 14 May 2008 from Mt. Jiajin, Baoxing County, Sichuan Province, China (30.683°N, 102.674°E; 2,855 m a.s.l.). Dried skin, cleaned skull, and alcohol-preserved carcass deposited in the

Kunming Natural History Museum of Zoology, KIZ.

Paratypes. KIZ-028550 (field ID: 0905186), adult female collected by Xuelong Jiang on 15 May 2008 from Mt. Jiajin, Baoxing County, Sichuan Province, China (30.680°N, 102.695°E; 2,785 m a.s.l.); KIZ-028551 (field ID: 0905198), adult male collected by Taiguang Huang on 15 May 2008 from Mt. Jiajin, Baoxing County, Sichuan Province, China (30.687°N, 102.696°E; 2,680 m a.s.l.). Cleaned skulls and alcohol-preserved carcasses were deposited in the Kunming Natural History Museum of Zoology, KIZ.

Measurements. See Supplementary Table S5.

Diagnosis. The largest species of *Uropsilus*. Skull significantly larger than that of all congeners and with laterally expanded zygomatic arches. Mandible robust and long; ML (14.58–14.86 mm) overlaps only with *U. investigator* and *U. andersoni* among congeners. Braincase distinctly broader than in *U. investigator*. Dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38, differing from *U. andersoni* (I 2/2, C 1/1, P 4/3, M 3/3 = 38). Dorsal pelage dark greyish-brown; ventral fur dark grey.

Description. Large *Uropsilus* species with body mass 10.35 ± 2.34 g (range 5.8–12.8 g) and HBL 75.17 ± 5.15 mm (range 65–80 mm). Ears large (EL = 9.67 ± 0.75 mm, range 8–10 mm), projecting above the pelage. Tail nearly as long as head and body, TL/HBL = 0.96 ± 0.04 (range 0.91–1.03); distal tail with greyish-brown bristle hairs ~ 10 mm long. Dorsal pelage greyish-brown; ventral fur dark grey. Dorsal and ventral pelage differ in color and are separated by a visible but not sharply contrasting boundary. Rump coloration matches dorsum. Dorsal surfaces of fore- and hindfeet and tail are dark brown. Medial surface of forefeet light brown, distinctly paler than hindfeet. Dorsal surface of tail dark brown; ventral surface paler brown.

Skull large: PL 22.14 ± 0.20 mm (range 21.95–22.48 mm), GNB 11.91 ± 0.19 mm (range 11.75–12.18 mm), ML 14.68 ± 0.09 mm (range 14.58–14.86 mm). In dorsal view, zygomatic arches markedly expanded laterally, exceeding greatest breadth of braincase. Temporal crests approach and nearly meet along the midline, forming an X-shaped pattern. Palatal breadth across M^2 (PPB) relatively wide (3.38 ± 0.08 mm, range 3.25–3.51 mm). Angle between anterior margin of coronoid process and mandibular body approximately 90–100°. Dentition: I 2/1, C 1/1, P 4/4, M 3/3 = 38. Upper canine slightly larger than P^1 but smaller than P^2. P^3 smallest upper premolar and slightly retracted lingually; P_3 equal to or smaller than P_1, both smaller than the lower canine.

Etymology. The specific epithet *maximus* is a Latin superlative meaning "largest" or "greatest," referring to the notably large body and skull size of this species, the largest known member of the genus.

Comparison. See Supplementary Material.

Distribution and ecology. *Uropsilus maximus* sp. nov. occurs at middle to high elevations in the mountains of western Sichuan, China. Its known range lies east of the Dadu River, from Jiuzhaigou County in the north to Mt. Erlang in Tianquan County in the south, at elevations of 2,200–3,100 m. It inhabits mixed coniferous–broadleaf forests and arrow-bamboo thickets. The distribution partially overlaps that of *U. soricipes*, which typically occurs at lower elevations (1,500–2,500 m) in evergreen broadleaf forests. Although the range of *U. maximus* sp. nov. is adjacent to that of *U. andersoni*, there are currently no confirmed records of sympatry between these species.

Remarks. *U. maximus* sp. nov. is most closely related to *U. cinereonates* sp. nov., from which it can be distinguished by its markedly larger body and skull size. For *U. maximus* sp. nov.: PL = 21.95–22.48 mm, GNB = 11.75–12.18 mm, ML = 14.58–14.86 mm; for *U. cinereonates* sp. nov.: PL = 20.21–20.55 mm, GNB = 10.40–10.94 mm, ML = 12.67–13.44 mm.

4.4.3. *Uropsilus wulingshanensis Li YX, Liu SY et Jiang XL, sp. nov.* (*Figs. 1, 6, 7; Supplementary Fig. S6, S7*)

Zoobank LSID: urn:lsid:zoobank.org:act:C442A0E5-7203-440F-A159-C1BB1B45A702.

Holotype. KIZ-042983 (field ID: 20140179), an adult female collected by Tao Wan on 24 August 2014 from Mt. Fanjing, Jiangkou County, Guizhou Province, China (27.911°N, 108.698°E; 2,093 m a.s.l.). Dried skin, cleaned but damaged skull, and alcohol-preserved carcass deposited in the Kunming Natural History Museum of Zoology, KIZ.

Paratype. SAF-10130, an adult male collected by Shaoying Liu in 2010 from Changyang County, Hubei Province, China (30.493°N, 110.805°E; 1,540 m a.s.l.). Dried skin, cleaned but damaged skull, and alcohol-preserved carcass deposited in the Sichuan Academy of Forestry (SAF).

Measurements. See Supplementary Table S5.

Diagnosis. Dorsal pelage uniformly dark brown; ventral fur dark grey. Tail shorter than head and body (TL/HBL < 1.0). Externally very similar to *U. dabieshanensis* and *U. huanggangensis* but with a narrower snout. Dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38. Maxilla and dentition indistinguishable from those of *U. gracilis*, but mandibular condylar process longer and more expanded distally.

Description. Body mass 8.1–9.0 g; HBL 72–75 mm. Tail shorter than head and body, TL/HBL 0.83–0.88; distal tail bearing bristles approximately 5–10 mm in length. Dorsal pelage brown with reddish-brown hair tips, overall uniform. Ventral surface deep slate grey with slight sheen; dorsal and ventral pelage differ in color. Forefeet dark brown; plantar surfaces of hindfeet jet black. Tail weakly bicolored: dorsal surface jet black, ventral surface brown, with modest contrast.

Rostrum narrow (breadth across snout, GBSn = 6.78–7.43 mm). Mandible length 13.64–14.02 mm. Condylar process elongated with distinctly enlarged distal end. Anterior margin of coronoid process forms an angle > 90° with mandibular body. Dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38. I^1 as wide as but longer than I^2. Upper canine larger than P^1; both smaller than P^2 but larger than P^3. P^3 very small and slightly inclined lingually.

Etymology. The specific epithet *wulingshanensis* is formed from "Wulingshan" (the Wuling Mountains) plus the Latin suffix *–ensis* ("from" or "belonging to"), referring to the Wuling Mountains, the broader mountain range that includes the type locality.

Comparison. See Supplementary Material.

Distribution and ecology. *Uropsilus wulingshanensis* sp. nov. is known from the Wuling Mountains east of the Wujiang River, including Changyang County in southwestern Hubei Province and Mt. Fanjing in Jiangkou County, northern Guizhou Province. It occurs at elevations of 1,500–2,100 m. Available data indicate that *U. wulingshanensis* sp. nov. is geographically isolated from *U. gracilis* to the west by the Wujiang River and from *U. funiushanensis* to the north by the Yangtze River.

Remarks. *U. wulingshanensis* sp. nov. differs from the closely related *U. atronates*, *U. fansipanensis*, and *U. gracilis* by its darker, deep brown pelage (vs. brighter reddish-brown in the latter three) and a more robust condylar process. It differs from *U. huanggangensis* by a narrower rostrum and P^3/P_3 smaller than the canines (vs. equal to or larger than the canines in the latter). It can be distinguished from *U. leptocranium* sp. nov. by its P^1 being smaller than the canine (vs. subequal in the latter).

This description is based on two specimens with varying degrees of damage to the skulls, limiting the completeness of the morphological characterization. However, genomic evidence strongly supports its status as a distinct species. Additional specimens are needed to refine its morphometric diagnosis.

4.4.4. *Uropsilus leptocranium Li YX et Jiang XL sp. nov.* (*Figs. 1, 6, 7; Supplementary Fig. S6*)

Zoobank LSID: urn:lsid:zoobank.org:act:E966D4AC-0189–4953-89FD-17D8BC611B54.

Holotype. KIZ-022457 (field ID: BC1110001), an adult female collected by Xuelong Jiang on 1 October 2011 from Mt. Wuding, Pingchuan Town, Binchuan County, Yunnan Province, China (25.874°N, 100.755°E; 3,194 m a.s.l.). Dried skin, cleaned skull, and alcohol-preserved carcass deposited in the Kunming Natural History Museum of Zoology, KIZ.

Paratypes. KIZ-022458 (field ID: BC1110002), adult collected by

Changzhe Pu on 1 October 2011 from Mt. Wuding, Pingchuan Town, Binchuan County, Yunnan Province, China (25.875°N, 100.753°E; 2,957 m a.s.l.); KIZ-022466 (field ID: BC1110202), adult collected by Changzhe Pu on 7 October 2011 from Mt. Wuding, Pingchuan Town, Binchuan County, Yunnan Province, China (25.885°N, 100.759°E; 2,808 m a.s.l.). Dried skins, cleaned skulls, and alcohol-preserved carcasses deposited in the Kunming Natural History Museum of Zoology, KIZ.

Measurements. See Supplementary Table S5.

Diagnosis. Small-bodied species with small skull, only slightly larger than *U. atronates* and *U. parvoniger* sp. nov. Skull among the most slender in the genus, characterized by a very narrow braincase (GNB 10.46–10.81 mm) and straight, laterally unexpanded zygomatic arches. Externally similar to *U. wulingshanensis* sp. nov. Dorsal pelage dark brown and ventral pelage dull dark grey; rump typically darker than head and back (similar to *U. parvoniger* sp. nov.). Tail distinctly bicolored dorsoventrally (vs. unicolored in *U. parvoniger* sp. nov.). Terminal tail bristles < 10 mm (vs. > 10 mm in *U. atronates*). Dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38; upper P^1 equal in size to upper canine (P^1 smaller than C in *U. wulingshanensis* sp. nov.).

Description. Small-sized species with body mass 7.88 ± 0.82 g (range 6.3–8.8 g) and HBL 68.89 ± 3.75 mm (range 63–75 mm). TL/HBL = 0.89 ± 0.04 (range 0.85–0.98). Terminal tail bristles generally < 10 mm. Dorsal pelage dark brown; rump usually darker than head and dorsum. Ventral pelage dull dark grey, without sheen, sometimes interspersed with a few brown hairs. Dorsal and ventral pelage differ in color, but the boundary is indistinct. Dorsal surface of tail dark brown, ventral surface paler brown.

Skull small with very narrow braincase: PL 20.57 ± 0.29 mm (range 20.14–21.00 mm), GNB 10.61 ± 0.14 mm (range 10.46–10.81 mm), ML 13.11 ± 0.26 mm (range 12.69–13.59 mm). Rostrum narrow (GBSn 7.15 ± 0.16 mm, range 6.75–7.35 mm). Zygomatic arches slender, straight, and not expanded laterally. Lateral profile of cranium essentially straight, lacking an anterior bulge in front of the braincase. Mandibular coronoid process gracile; condylar process very short. Dentition: I 2/1, C 1/1, P 4/4, M 3/3 = 38. I1 longer and wider than I2. Upper canine equal in size to P^1, both smaller than P^2 but larger than P^3, which is slightly inclined lingually. In the mandible, only I_2 is present and is enlarged and chisel-shaped. P_1 slightly larger than P_3; both smaller than the lower canine. Lower canine, P_2, and P_3 increase in size sequentially.

Etymology. The specific epithet *leptocranium* is derived from the Greek *leptos* (slender) and *kranion* (skull), referring to the distinctly slender skull and straight, laterally unexpanded zygomatic arches. The epithet is treated as a noun in apposition and is therefore invariable.

Comparison. See Supplementary Material.

Distribution and ecology. *Uropsilus leptocranium* sp. nov. is currently known from Mt. Wuding in Binchuan County and the nearby Mt. Maotai in Dayao County, Yunnan Province. It inhabits evergreen broadleaf forests and mixed coniferous–broadleaf forests at ~ 2,800–3,400 m, typically in moss-covered dwarf forests near mountain summits.

Remarks. *U. leptocranium* sp. nov. differs from the closely related *U. atronates* by its deep brown dorsal pelage, indistinct dorsoventral color boundary, and shorter terminal tail bristles (< 10 mm), whereas *U. atronates* has reddish-brown dorsal pelage, a sharply defined dorsoventral color boundary, and longer terminal tail bristles (10–20 mm). It can be distinguished from *U.* cf. *fansipanensis* by its darker dorsal pelage and a notably narrower braincase.

4.4.5. *Uropsilus ningshanensis Li YX et Jiang XL, sp. nov.* (Figs. 1, 6, 7)

Zoobank LSID: urn:lsid:zoobank.org:act:380AC325-8EA2-4B42-A7E1-8C499828F1F6.

Holotype. KIZ-016111 (field ID: 0509576), an adult collected by Xuelong Jiang on 23 September 2005 from eastern Qinling, Guanghuojie Town, Ningshan County, Shaanxi Province, China (33.760°N, 108.771°E; 1,190 m a.s.l.). Dried skin, cleaned skull, and alcohol-preserved carcass deposited in the Kunming Natural History Museum of Zoology, KIZ.

Paratypes. KIZ-016100 (field ID: 0509337), adult collected by Xuelong Jiang on 15 September 2005 from eastern Qinling, Xunyangba Town, Ningshan County, Shaanxi Province, China (33.495°N, 108.521°E; 1,706 m a.s.l.); KIZ-016107 (field ID: 0509513), adult collected by Xuelong Jiang on 21 September 2005 from eastern Qinling, Guanghuojie Town, Ningshan County, Shaanxi Province, China (33.836°N, 108.800°E; 1,978 m a.s.l.). Dried skins, cleaned skulls, and alcohol-preserved carcasses deposited in the Kunming Natural History Museum of Zoology, KIZ.

Measurements. See Supplementary Table S5.

Diagnosis. Distinguished from all congeners except *U. soricipes* by having the minimal dental count: I 2/1, C 1/1, P 3/3, M 3/3 = 34, lacking both upper and lower P3. General external and cranial morphology very similar to *U. soricipes*, but dorsal pelage uniformly dark reddish-brown with redder hair tips than in *U. soricipes*; ventral pelage dark grey. Anterior margin of mandibular coronoid process forms an angle ≤ 90° with mandibular body (vs. > 90° in *U. soricipes*).

Description. Medium-sized *Uropsilus* species with body mass 9.40 ± 1.71 g (range 8–13 g) and HBL 69.74 ± 4.71 mm (range 59–78 mm). Ears very small (EL = 7.47 ± 0.68 mm, range 6–9 mm), pinnae slightly projecting above the pelage. TL/HBL = 0.86 ± 0.07 (range 0.78–1.05); terminal tail bristles brown, 5–10 mm long. Dorsal pelage dark brown; ventral pelage dark grey with scattered brownish-yellow tipped hairs. Dorsal–ventral color contrast present but boundary indistinct. Feet light brown, with no obvious color difference between fore- and hindfeet. Tail bicolored: dorsal side dark brown, ventral side light brown.

Skull measurements: PL 20.91 ± 0.37 mm (range 20.04–21.48 mm), GNB 11.32 ± 0.22 mm (range 10.79–11.65 mm), ML 13.54 ± 0.25 mm (range 13.06–14.03 mm). Temporal ridges prominent and converging medially on braincase. Braincase full and slightly elevated anteriorly in lateral view. Anterior margin of coronoid process forms an angle ≤ 90° with mandibular body. Dental formula: I 2/1, C 1/1, P 3/3, M 3/3 = 34. I^1 longer and wider than I^2. Upper canine and P^1 nearly equal in size, both smaller than P^2. P^3 completely absent. Mandible with only a second lower incisor (I_2), enlarged and chisel-shaped. P_1 smaller than lower canine; P_3 absent. Lower canine, P_2, and P_4 increase in size sequentially.

Etymology. The specific epithet *ningshanensis* is formed from "Ningshan", the type locality, plus the Latin suffix *−ensis*, meaning "from" or "of," referring to Ningshan County where this species was discovered.

Comparison. See Supplementary Material.

Distribution and ecology. *Uropsilus ningshanensis* sp. nov. is currently known only from eastern Qinling, Ningshan County, Shaanxi Province. It inhabits evergreen broadleaf and bamboo forests at mid elevations of 1,300–2,500 m.

Remarks. *Uropsilus ningshanensis* sp. nov. is most readily distinguished from the closely related *U. soricipes* by the shape of the mandible: the anterior margin of the coronoid process forms an angle ≤ 90° with the mandibular body (vs. > 90° in *U. soricipes*).

4.4.6. *Uropsilus cinereonates Li YX, Song WY et Jiang XL sp. nov.* (Figs. 1, 6, 7)

Zoobank LSID: urn:lsid:zoobank.org:act:DD30C938-A011-48EF-8256-EC7FB3C750D8.

Holotype. KIZ-042984 (field ID: PS2306035), an adult female collected by Mingjin Pu on 29 June 2023 from Chuanzhusi Town, Songpan County, Sichuan Province, China (32.462°N, 103.395°E; 3,232 m a.s.l.). Dried skin, cleaned skull, and alcohol-preserved carcass deposited in the Kunming Natural History Museum of Zoology, KIZ.

Paratypes. KIZ-042985 (field ID: PS2306036), adult male collected by Mingjin Pu on 29 June 2023 from Chuanzhusi Town, Songpan County, Sichuan Province, China (32.462°N, 103.395°E; 3,232 m a.s.l.); KIZ-042986 (field ID: PS2306073), adult female collected by Wemyu

Song on 2 July 2023 from Chuanzhusi Town, Songpan County, Sichuan Province, China (33.349°N, 103.435°E; 3,328 m a.s.l.). Dried skins, cleaned skulls, and alcohol-preserved carcasses deposited in the Kunming Natural History Museum of Zoology, KIZ.

Measurements. See Supplementary Table S5.

Diagnosis. Dorsal pelage deep grey to grey-brown. Colored portion of dorsal hair tips minimal and pale, appearing as light yellow restricted mainly to head and anterior dorsum; rump and ventral surfaces uniform deep grey. Dorsal and ventral pelage lack a distinct boundary. Tail distinctly bicolored dorsoventrally, with ventral surface noticeably paler than dorsal. Tail elongate, exceeding head–body length (TL/HBL > 1.0). Differs from all congeners except *U. parvoniger* sp. nov. (tail unicolored dorsoventrally), *U. leptocranium* sp. nov. (tail shorter than body), and *U. atronates* (dorsal hair tips bright chestnut-red with a distinct dorsoventral color boundary) in having conspicuously smaller body and cranial dimensions, shorter tooth rows, and a more slender mandibular coronoid process. Dental formula typically I 2/1, C 1/1, P 4/4, M 3/3 = 38, but upper and lower P3 show variable reduction and may be absent.

Description. Small-sized species with body mass 8.60 ± 1.05 g (range 7.60–10.3 g) and HBL 68.75 ± 4.15 mm (range 65–75 mm). Tail exceeds head–body length, TL/HBL 1.05 ± 0.04 (range 1.00–1.09). Tail tip bears short grey bristles. Dorsal pelage overall dark grey or brownish-grey with scattered pale yellowish hair tips. Head has the highest density of pale tips, which fade posteriorly towards the rump, where dorsal coloration becomes similar to ventral deep grey. Dorsal and ventral pelage nearly continuous in tone, but tail is distinctly bicolored dorsoventrally.

Skull small: PL 20.36 ± 0.15 mm (range 20.21–20.55 mm), GNB 10.76 ± 0.21 mm, ML 13.11 ± 0.32 mm (range 12.67–13.44 mm). Nasal rostrum short (MPL 9.10 ± 0.23 mm, range 8.82–9.42 mm). Zygomatic arches slender and not obviously expanded laterally. Infraorbital foramen similar in size to lacrimal foramen. Mandibular body and coronoid process very gracile; anterior margin of coronoid process approximately perpendicular to mandibular body. Dentition: I 2/1, C 1/1, P 4/4, M 3/3 = 38, but upper P^3 and lower P_3 are variably reduced or absent. I^1 and I^2 equal in width, with I^1 longer. Upper canine equal in size to P^1, both smaller than P^2. In the mandible, only one incisor (I_2) is present, chisel-shaped. P_1 distinctly larger than P_3 (if present), both smaller than the lower canine.

Etymology. The specific epithet *cinereonates* is derived from the Latin *cinereus* (ash-grey) and *nates* (rump), referring to the overall greyish coloration and especially the dark grey rump, which contrasts with the more brownish tones of many congeners. The epithet is treated as a noun in apposition and is therefore invariable.

Comparison. See Supplementary Material.

Distribution and ecology. *Uropsilus cinereonates* sp. nov. is currently known only from Songpan County, Sichuan Province. It inhabits larch-dominated forests at elevations of 3,200–3,400 m. Its distribution partially overlaps with those of *U. soricipes* and *U. maximus* sp. nov., but the latter two species occur at lower elevations (1,500–2,500 m and 2,200–3,100 m, respectively).

Remarks. *Uropsilus cinereonates* sp. nov. is readily distinguished from its closest relative, *U. maximus* sp. nov., by its markedly smaller body and cranial size. For *U. cinereonates* sp. nov.: PL = 20.21–20.55 mm, GNB = 10.40–10.94 mm, ML = 12.67–13.44 mm; for *U. maximus* sp. nov.: PL = 21.95–22.48 mm, GNB = 11.75–12.18 mm, ML = 14.58–14.86 mm.

4.4.7. Key to the species of the genus Uropsilus (17 species)

1a. 36 teeth; dental formula I 2/2, C 1/1, P 3/3, M 3/3 = 36	***U. aequodonenia***
1b. 34 or 38 teeth	2
2a. 34 teeth; dental formula I 2/1, C 1/1, P 3/3, M 3/3 = 34	3
2b. 38 teeth	4
3a. Anterior margin of coronoid process forming an angle > 90° with mandibular body; mountains of western Sichuan	***U. soricipes***
3b. Anterior margin of coronoid process nearly vertical (angle ≤ 90°) to mandibular body; Qinling region, Shaanxi Province	***U. ningshanensis* sp. nov.**
4a. Two lower incisors; third lower premolar absent; dental formula I 2/2, C 1/1, P 4/3, M 3/3 = 38	***U. andersoni***
4b. One lower incisor; third lower premolar present; dental formula I 2/1, C 1/1, P 4/4, M 3/3 = 38	5
5a. Pelage bright with slight sheen; dorsal fur yellow–brown or reddish-brown, with a distinct boundary between dorsal and ventral pelage	6
5b. Pelage dull, without obvious sheen; dorsal fur black, grey, or dark brown; dorsal and ventral coloration similar or with an indistinct boundary	10
6a. Dorsal hair tips reddish to chestnut-brown	7
6b. Dorsal hair tips glossy blackish-gold or smoky yellow	9
7a. Ventral pelage uniform, slightly glossy dark slate-grey; terminal caudal bristles 10–20 mm long	***U. atronates***
7b. Ventral pelage dark grey, mixed with a few yellow–brown hairs; terminal caudal bristles ≤ 10 mm	8
8a. Lower third premolar (P_3) ≥ lower first premolar (P_1) in size	***U. fansipanensis***
8b. Lower third premolar (P_3) < lower first premolar (P_1) in size	***U. gracilis***
9a. Dorsal hair tips blackish-gold; ventral pelage pale slate-grey	***U. nivatus***
9b. Dorsal hair tips smoky yellow; ventral pelage dark grey	***U. funiushanensis***
10a. Tail unicolored dorsally and ventrally (blackish or dark brown)	11
10b. Tail distinctly bicolored dorsoventrally, ventral surface paler than dorsal surface	12
11a. Mandible length > 13.80 mm	***U. investigator***
11b. Mandible length ≤ 13.80 mm	***U. parvoniger* sp. nov.**
12a. Greatest breadth of snout (GBSn) 7.47–8.68 mm	13
12b. Greatest breadth of snout (GBSn) 6.75–7.43 mm	15
13a. Tail length > 90% of head–body length (TL/HBL > 0.90)	***U. maximus* sp. nov.**
13b. Tail length ≤ 90% of head–body length (TL/HBL ≤ 0.90)	14
14a. Third upper and lower premolars (P3) equal to or larger than canines in size	***U. huanggangensis***
14b. Third upper and lower premolars (P3) distinctly smaller than canines	***U. dabieshanensis***
15a. Tail length ≥ 100% of head–body length (TL/HBL ≥ 1.00)	***U. cinereonates* sp. nov.**
15b. Tail length < 100% of head–body length (TL/HBL < 1.00)	16
16a. First upper premolar (P^1) equal in size to upper canine (C)	***U. leptocranium* sp. nov.**
16b. First upper premolar (P^1) smaller than upper canine (C)	***U. wulingshanensis* sp. nov.**

CRediT authorship contribution statement

Yi-Xian Li: Visualization, Writing – original draft, Writing – review & editing, Formal analysis, Methodology, Investigation. **Quan Li:** Writing – review & editing, Investigation, Resources, Methodology. **Wen-Yu Song:** Writing – review & editing, Investigation, Resources. **Xue-You Li:** Writing – review & editing. **Kenneth Otieno Onditi:** Writing – review & editing. **Xiao-Xin Pei:** Methodology. **Hong-Jiao Wang:** Data curation. **Chang-Kun Fu:** Resources. **Shao-Ying Liu:** Resources. **Zhong-Zheng Chen:** Writing – review & editing, Supervision, Investigation, Resources. **Xue-Long Jiang:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition, Project administration, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ympev.2026.108652.

Data availability

All raw whole-genome resequencing data used in this study have been deposited in the NCBI Sequence Read Archive (SRA) under BioProject accession PRJNA1187160. The samples newly generated in this study correspond to BioSample accession SAMN53512958-SAMN53512962.

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